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Fast Analysis Method of Diagnostic Fracture Injection Test Data: Examples from Four Different Shale Formations (Bakken, Three Forks, Wolfcamp, and Eagle Ford)

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Abstract

This study introduces a new methodology for analyzing DFIT data by applying Gaussian Pressure Transient (GPT) solutions to model pressure fall-off behavior in hydraulically fractured wells. Diagnostic Fracture Injection Test (DFIT) data from 7 wells were analyzed; the wells were drilled in four different shale formations (Wolfcamp, Eagle Ford, Bakken, and Three Forks). All wells analyzed were hydraulically fractured, and as a first step in the fracturing treatment operation, each of the completion companies involved in the different wells applied a DFIT test to determine the ISIP and the formation leak-off rate to estimate the permeability. The original pressure fall-off excel sheets, availed by different operators, were analyzed in detail. For each test, the relevant part of the leak-off test data was identified and isolated for further in-depth study of the leak-off rate. Instead of relying on traditional DFIT interpretation techniques such as G-function or Carter leak-off assumptions, the method leverages the Gaussian diffusion model to capture the pressure propagation and leak-off characteristics more accurately. The Gaussian pressure transient analysis combines pressure and volume balancing principles directly on the actual field measurements. The results show that the Gaussian method successfully matched the pressure fall-off curves for all 7 wells, with improved accuracy in estimating leak-off rates and hydraulic diffusivity. Moreover, the GPT method requires only a short-term pressure fall-off curve to accurately determine reservoir properties, and this can significantly shorten the typical long fall-of time in low permeability rocks. The Gaussian pressure model provided excellent matches with field data, confirming the method's reliability for unconventional reservoir analysis. The proposed methodology thus offers significant advancement over conventional techniques, better aligning DFIT analysis with the complex realities of unconventional reservoirs.

Introduction

Diagnostic Fracture Injection Tests (DFITs) have become a critical tool for reservoir characterization, particularly in unconventional resource plays where traditional well testing is impractical. By analyzing pressure fall-off behavior after a small volume of controlled fluid injection, DFITs provide essential parameters such as in-situ stress, leak-off rate, and reservoir permeability, which are fundamental for optimizing the subsequent hydraulic fracturing design and field development strategies.

Over the past decades, a range of traditional DFIT interpretation techniques have been developed, including G-function analysis, square-root-of-time plotting, and classical pressure transient models based on linear and radial flow assumptions. While these methods offer practical workflows, they rely on simplifying assumptions such as ideal Carter leak-off behavior and instantaneous fracture closure which often do not hold true in complex, low-permeability reservoirs. Consequently, interpretation accuracy can suffer, particularly when faced with phenomena like gradual fracture closure, pressure-dependent leak-off rate, and multiphase flow. The present study introduces a new Gaussian pressure transient solution method for accurate leak-off quantification in combination with pressure and volume balancing field test data.

Before applying the newly proposed method in detailed case studies, this introduction first provides an overview of DFIT operations and objectives (Section 1.1), briefly reviews the traditional interpretation techniques (Section 1.2) and prior analytical methods for modeling DFIT data (Section 1.3), and outlines the new Gaussian pressure transient modeling methods for DFIT applications (Section 1.4).

Overview of DFIT and its objectives

In a Diagnostic Fracture Injection Test (DFIT), a small volume of fluid is injected into a well to intentionally create a single hydraulic fracture of limited length; next, the well is shut-in and pressure fall-off is recorded for hours or days (McClure, 2017). DFITs are critical in reservoir characterization of low-permeability (unconventional) formations where conventional well tests are impractical. By analyzing the pressure fall-off, engineers can infer key formation and stress parameters. Important DFIT information gathering targets are: (1) the instantaneous shut-in pressure (ISIP) – the pressure just after injection stops (indicative of the pressure in the fracture at shut-in), (2) the fracture closure pressure – an estimate of the minimum in-situ stress (when the fracture closes), and (3) the fluid leak-off rate into the formation. DFIT analysis also yields estimates of initial reservoir (pore) pressure and permeability, especially from late-time data after fracture closure (McClure, 2017). These parameters (stress, pressure, permeability, leak-off) are vital for executing the subsequent full hydraulic fracturing treatment in an effective fashion.

Why DFIT is such an important diagnostic test? The answer is that, in ultra-tight shales, performing a standard pressure transient test without fracturing is impossible – fluid injection at meaningful rates will not directly leak-off into the formation as would occur in conventional formations of high permeability. Instead, an hydraulic fracture will be created due to the shale's ultra-low permeability. DFIT provides an in-situ measurement using a mini-frac test under actual stress conditions that economically yields essential data (stress profile, reservoir pressure, etc.) before full-scale stimulation (Jacobs, 2019). The outcomes guide completion design (e.g. fracture treatment pressures) and well spacing decisions by quantifying how the formation will fracture and flow.

Several studies have advanced the state of DFIT data interpretation using combinations of analytical, numerical, and field-based approaches. In a large collaborative study, McClure et al. (2019), integrated computational modeling, field data, and synthetic cases using three-dimensional fracture, reservoir, and wellbore simulation to assess DFIT interpretation protocols. These analyses explored the effects of changing fracture compliance and system storage, revealing that commonly used Carter-based or G- function methods can underpredict permeability and fail to capture complex field behaviors in unconventional shales. The study led to refined protocols for multi-well DFIT interpretation, applied across a variety of North American shale plays.

Wang and Sharma (2019) emphasized the limitations of conventional assumptions, highlighting how both Carter's leak-off model and the premise of constant fracture compliance are violated during fracture closure, especially in low-permeability reservoirs. Their analysis revealed that traditional methods cannot consistently fit DFIT pressure decline data before and after closure. They proposed a modified analytical framework to address these shortcomings.

Du and Xu (2024) implemented a fully coupled poroelastic finite element model to simulate fracture initiation, propagation, and closure, explicitly addressing leak-off coefficient estimation and in-situ stress. Results from this study demonstrated that incorporating poroelastic effects could yield leak-off coefficients up to double those derived from previous (Carter-based) techniques, underscoring the role of coupled geomechanical responses in DFIT data interpretation.

Analytical advances have also been proposed; (Cai et al., 2020) developed a material-balance and superposition-based approach to estimate formation permeability, accounting for fluid leak-off during both fracture propagation and shut-in periods.

Attempts to expedite DFIT analysis in operational settings have produced alternative protocols such as DFIT-FBA (flowback analysis), which can dramatically reduce test time (Zeinabady and Clarkson, 2021) Haqparast et al. (2024) applied the novel method to unconventional reservoirs, introducing corrections for operational effects like perforation friction and capillary pressure. However, these approaches still rely on conventional leak-off assumptions and do not incorporate advanced pressure transient-diffusion (e.g., Gaussian) solutions.

Oshaish and Weijermars (2023) introduced a Sneddon-based analytical model for stepwise, dynamic estimation of fracture half-length and propagation using practical spreadsheet tools and validated on field data from the Eagle Ford shale. This facilitates a more direct estimation of fracture growth and closure in a manner compatible with post-injection DFIT analysis. Building on their 2023 work (Weijermars & Oshaish, 2023), Weijermars and Oshaish (2025) presented a fast-acting approach for inferring fracture half-length and propagation rates from post-fracturing data by integrating Sneddon's pressure theory with dynamic fracture width and volumetric mass balance. This method showed strong fidelity and is highly relevant for DFIT and mini-frac test interpretation.

Traditional DFIT interpretation techniques

Traditional DFIT interpretation relies on specialized pressure-decline analysis methods originally developed for mini-frac tests in conventional reservoirs. The most common techniques include Nolte's G-function analysis, the square-root-of-time plot, and related methods introduced by Nolte and others in the 1980s. Each method has underlying assumptions and known limitations:

- **G-function analysis (Nolte's Method):** Nolte (1979) introduced the G-function as a modified time function to interpret leak-off behavior during and after injection (Whitson, n.d.). It is derived assuming Carter's leak-off model (fluid leak-off into the formation at a rate proportional to √time) for a planar fracture (Gabry et al., 2024). In practice, one plots pressure vs. G-time and its derivatives. Fracture closure is identified by a change in slope of the G*dP/dG curve – classically, when the derivative trend deviates upward, indicating the fracture stiffens as walls begin to contact. Nolte's method also classifies leak-off regimes (e.g. normal vs. abnormal leak-off) based on the shape of this curve. Assumptions: a single planar fracture of constant height, no proppant, and that fracture growth stops immediately at shut-in (Gabry et al., 2024). These ensure a simple relationship between leaked volume and time (the basis of the G-function).
- **Square-root-of-time plot:** This is a graphical technique to pick closure pressure by plotting pressure decline against the square root of shut-in time. Prior to fracture closure, if fluid leak-off is dominated by one-dimensional (linear) flow from the fracture, the pressure vs. √t trend appears linear. Extrapolating this linear trend and finding the point where the actual pressure curve departs from it gives the closure pressure (the "tangent" method). In essence, when the fracture is open, pressure fall- off follows a √t linear slope, and once the fracture closes (flow regime changes), the curve diverges. Assumptions: early-time linear flow and a clear, singular closure event causing a distinct change in slope. Often, the square-root plot is used alongside the G-function as a confirmation of the picked closure stress (Gabry et al., 2024).

- **Nolte's other analytical methods:** In addition to the G-function, Nolte and Smith (1981) and others proposed log-log diagnostic plots and type-curves for minifrac analysis. For example, a log-log plot of pressure drops and its derivative versus time can reveal flow regimes: a 1/2-slope line before closure (signature of linear flow from an open fracture) and flattening derivative after closure (transition to radial flow) (Gabry et al., 2024). Nolte's overall approach combined before-closure analysis (BCA) to determine closure and fluid loss type, and after-closure analysis (ACA) to estimate permeability from late-time data. Underlying these methods are the same simplifying assumptions (single fracture, Carter leak-off, etc.), which enabled analytical solutions for pressure decline.

Limitations. These traditional techniques, while widely used, have well-documented limitations. They assume ideal leak-off behavior (Carter's model) and a prompt, distinct fracture closure. In reality, these assumptions can break down. For instance, the G-function method assumes constant-rate leak-off and no fracture growth after shut-in (Gabry et al., 2024) – conditions that may be violated if the fracture continues to propagate briefly or if leak-off rate changes with pressure. The G-function pick for permeability relies on a single point (the slope at closure), which can be skewed by near-wellbore effects (e.g. tortuosity or pressure drop in the wellbore). Moreover, Carter leak-off may no longer be valid long after shut-in as pressure falls, meaning the G-time transform itself becomes approximate. The square- root-time method (tangent method) is noted to be sensitive to post-closure fluid dynamics– if, for example, gradual closure or changing compressibility affects the decline, the deviation point can be ambiguous. In low-permeability shales, fracture closure is not an instantaneous event but a gradual process, which can smear out the "corner" on a √t plot or G-function plot. Thus, classical plots sometimes yield ambiguous or subjective closure picks. In summary, traditional DFIT interpretation techniques are straightforward but hinge on idealized behavior. Their assumptions (single-phase flow, constant fracture stiffness, etc.) can lead to errors when applied to real field data that deviate from these ideal conditions (Gabry et al., 2024).

Barree et al. (2009) introduced a comprehensive methodology for interpreting pre-fracture diagnostic injection tests (DFITs) by integrating multiple analytical techniques to enhance the accuracy of fracture diagnostics. Recognizing the limitations and potential misinterpretations associated with relying solely on traditional methods like the square-root time plot, the authors advocated for a combined approach utilizing G-function analysis, its derivatives, and log-log pressure derivative plots. Through the examination of four field case studies, they demonstrated how this integrated analysis could consistently determine critical parameters such as closure pressure and time, as well as identify pre- and post-closure flow regimes and reservoir properties. This holistic approach addresses the complexities inherent in various leak-off behaviors and fracture storage scenarios, providing a more reliable framework for DFIT interpretation and subsequent hydraulic fracture design.

Analytical pressure-transient modeling of DFIT data

After fracture closure, the DFIT essentially becomes a pressure fall-off test in the formation, and classical pressure transient analysis techniques can be applied to glean reservoir properties. The pressure decline can be interpreted in terms of flow regimes that are analogous to those in traditional well testing (injection/falloff). Classical analytical models for transient flow – notably linear flow and radial flow – are used to describe different periods of the DFIT pressure-time curve:

- **Linear flow (pseudo-linear after closure):** Immediately after the fracture closes, the remaining fluid in the fracture must dissipate into the formation. This is often approximated as one-dimensional flow linear flow into the reservoir, because the fracture had a roughly planar shape. During this period, pressure decline vs. time$^{0.5}$ tends to follow a straight line (hence the √t plot usefulness) corresponding to linear diffusion from a line-source. In log-log diagnostic terms, a half-slope (0.5) line on a pressure derivative plot is characteristic of linear flow regime. The linear flow regime is used to infer the contact area of the fracture and leak-off characteristics. In practice, engineers sometimes match early after-closure data to linear flow analytical solutions to estimate an effective fracture half-length or transmissivity of the fracture face.

- **Radial flow (pseudo-radial late-time):** Given sufficient time, the pressure transient will eventually enter a radial flow regime, where the pressure fronts move spherically outward from the well, as if from a point source. In a DFIT, once the fracture is fully closed and the injected fluid has mostly leaked off, the wellbore and formation behave like a small-radius injection that has ceased – analogous to a shut-in after a small injection. If the test is long enough, one can observe a leveling of the log-log derivative (flat derivative) or a classic semilog straight-line on a Horner plot, indicating radial flow in the formation. Identifying a radial flow period is extremely useful: it allows determination of reservoir permeability and initial pressure using standard well test analysis methods (e.g. semilog slope = $m = 162.6\ q\mu/kh$, and intercept for reservoir pressure). In DFIT practice, engineers look at late-time data for a pseudo-radial flow signature (for instance, a unit slope line on a log-log plot transitioning to 0 slope derivative) (Gabry et al., 2024). If present, this provides a direct means to compute permeability and pore pressure. In fact, DFIT interpretation workflows explicitly include checking the after-closure period for linear → radial flow regime transition. The after-closure analysis (ACA) uses these flow regimes: a linear flow model may be fit to early after-closure data, and a radial flow model to later data, to jointly solve for formation $k\,\phi$.

These analytical models derive from solutions to the diffusivity equation (pressure diffusion in porous media). For example, the radial flow solution is analogous to the Theis solution or classic well test exponential decline, and linear flow can be described by analytical solutions for flow from a line source (fracture face) into an infinite medium. By matching DFIT pressure decline to these solutions (often via log-log type-curve matching or specialized plots), one can extract parameters: permeability, pore pressure, fracture surface area, etc. A key point is that both flow regimes may not always be observed in every DFIT. In very tight formations, radial flow may only occur after days or weeks, often beyond the duration of the test. In those cases, analysis might rely on linear flow only, yielding a range or lower bound of permeability. Nonetheless, pressure transient modeling complements the traditional DFIT plots by leveraging well test theory: the early portion of the fall-off is treated like a linear transient from a finite fracture, and the late portion like a radial well test. Modern DFIT interpretation software will generate both G-function plots and log-log derivative plots to integrate these classical analytical interpretations (Gabry et al., 2024) (Whitson, n.d.). This pressure transient approach grounds DFIT analysis in solid reservoir engineering principles (radial/linear flow solutions), but still faces the challenges of unconventional effects (e.g. identifying a clear radial flow can be difficult if multiphase phenomena dominate or test time is short).

Gaussian Pressure Transient modeling

In recent years, a novel approach termed Gaussian pressure transient modeling has been proposed as an alternative analytical method for interpreting pressure data. Pioneered by Weijermars and colleagues, GPT solutions offer a fresh perspective on the pressure diffusion problem by using Gaussian analytical functions to describe pressure propagation in the reservoir. In essence, GPT modeling provides closed- form solutions to the diffusivity equation that capture the pressure response to injection or production as a Gaussian distribution in space and time (Weijermars, 2022).

Concept. Weijermars (2022) introduced new pressure transient solutions in which an instantaneous pressure perturbation from a source (wellbore or fracture) spreads out in the reservoir following a Gaussian profile. For example, consider a planar fracture as a pressure source: the GPT solution describes how the pressure front emanates from that plane into the reservoir with time, using Gaussian functions that satisfy the diffusion equation for a particular well completion arrangement. A remarkable feature of GPT solutions is that they do not require the well's flow rate as an explicit input – instead of prescribing rate and solving for pressure (as in conventional well testing), GPT directly solves for the pressure field due to an imposed pressure on the well system, and then derives the flux from the pressure gradient via Darcy's law. In practical terms, this means one can model a shut-in or production decline without having to piecewise assume constant rates; the solution inherently accounts for the pressure-change propagation. The Gaussian formulation acts almost like a Green's function for the reservoir: any injection/falloff scenario can be constructed by superposition of these basic solutions.

Recent Contributions. Weijermars and co-authors have applied GPT modeling to a variety of reservoir engineering problems. In production decline analysis, Gaussian Decline Curve Analysis has been used to history-match production data with notable success. For instance, Weijermars and Oshaish (2023) used a GPT-based method to estimate hydraulic fracture half-lengths by matching post-frac production, achieving results consistent with microseismic measurements and traditional rate-transient analysis. This demonstrated that GPT models can reliably capture the effects of fracture geometry on pressure transients. Similarly, Oshaish and Weijermars (2025) developed a new stepwise modeling approach (based on Gaussian transients) to track fracture propagation during pump-in, showing that field pumping data could be reproduced stage-by-stage with this method. These works indicate the broad utility of GPT solutions – from interpreting flow regimes to estimating fracture dimensions – due to their analytical nature and fast convergence. In the context of DFIT, GPT provides a way to analyze the entire pressure fall-off curve holistically. Rather than splitting the analysis into pre-closure vs. post-closure with different models, a Gaussian model could, in principle, model the pressure decline from the moment of shut-in through to late-time with a single continuous solution.

Advantages over Traditional Methods. GPT-based modeling offers several potential improvements for DFIT interpretation. Firstly, because it is rooted in fundamental solutions of the diffusion equation, it can handle arbitrary injection histories via superposition – this relaxes the need for strictly constant-rate injection assumptions (a known limitation of G-function analysis which formally assumes a constant-rate prior to shut-in). (Gabry et al., 2024). Secondly, the closed-form nature of GPT means it is computationally efficient and can be implemented in simple tools (even spreadsheets) for quick analysis (Weijermars, 2022). This is in line with industry's desire for fast, practical DFIT interpretation methods that do not require heavy numerical simulation. Thirdly, GPT provides a different diagnostic viewpoint: for example, a mis-match between a Gaussian analytical pressure decline and field DFIT data might indicate the influence of phenomena not captured (e.g. multi-phase leak-off), thereby highlighting those effects. Early studies suggest GPT can match field pressure decline data well; Weijermars' team reported that GPT-derived estimates of fracture properties were in close agreement with those obtained from conventional analyses and even direct observations. In summary, Gaussian pressure transient modeling is an emerging analytical framework that complements and, in some ways, modernizes DFIT analysis. By treating the pressure fall-off as a diffusion process that can be described by Gaussian kernel solutions, it provides a novel perspective – one that potentially avoids some pitfalls of the older methods and offers a more direct handle on the physics of pressure propagation from fractures.

Research gap and need for a new approach

The literature review above highlights a clear gap between conventional DFIT interpretation methods and the complex reality of unconventional reservoirs. Traditional techniques (G-function plots, √t plots, etc.) were developed decades ago for relatively simple scenarios and often struggle with modern DFIT data, which involve phenomena like variable fracture compliance, pressure-dependent leak-off, and multi-phase flow. Despite improvements – such as the compliance method for picking closure or extended after-closure analysis for permeability – practitioners still face ambiguity in analysis. Field studies have revealed systematic biases (e.g. over-estimated permeability in shale gas DFITs using classical methods. and cases where standard plots fail to yield a definitive result (no clear closure signal, etc.). This suggests that existing interpretation workflows are reaching their limits in the face of unconventional reservoir complexity.

Researchers and engineers have recognized the need for new approaches to DFIT interpretation that are more robust and physically representative. One direction has been to incorporate more physics into models – for example, Wang and Sharma (2017) introduced a variable compliance DFIT model accounting for gradual fracture closure and unsteady leak-off, which significantly improved stress estimation accuracy. Another direction is to gather more data during DFITs: Clarkson and others have advanced the DFIT-FBA (flowback analysis) method, where after the shut-in period, a controlled flowback is performed to collect additional pressure data, thereby capturing multi-phase flow effects and accelerating the test timeline. These innovations address specific shortcomings (closure pick, multiphase leak-off), but they also add complexity to the testing or interpretation procedure.

Amid these developments, the Gaussian Pressure Transient (GPT) modeling emerges as a promising new analytical approach. The GPT framework can be seen as addressing the research gap by providing a unified, physics-based description of the pressure fall-off. It leverages the diffusion equation in its pure form, potentially incorporating various effects through its parameters (e.g. an effective hydraulic diffusivity that could reflect fluid properties and rock permeability). Because GPT solutions are superposable, they can naturally handle variable injection schedules or even piecewise changes in leak- off behavior without switching models. This continuous modeling could improve the characterization of the entire DFIT pressure decay curve, capturing early-time and late-time behavior in one sweep. Early research by Weijermars and co-workers has demonstrated GPT's capability in related applications (production decline, fracture sizing) with accurate results, hinting that similar success could be achieved for DFIT interpretation. In particular, a GPT-based DFIT analysis might better honor the actual physics (e.g. gradual closure can be represented by a gradual change in pressure diffusion regime, rather than an abrupt break in model). Moreover, GPT models are typically fast and algebraic, lending themselves to probabilistic or iterative analyses (such as the probabilistic fracture length estimation done by Alvayed et al. (2023) using GPT. This means field engineers could iterate on DFIT results quickly, exploring uncertainties in a practical way.

In summary, the research gap lies in the inability of conventional DFIT interpretations to reliably handle non-ideal behaviors prevalent in unconventional reservoirs. There is a need for interpretation methodologies that incorporate complex leak-off, fracture mechanics, and multi-phase fluid effects without overly relying on idealized assumptions. GPT-based models represent a compelling new approach to fill this gap. By providing a novel analytical toolkit grounded in the fundamental diffusion physics, GPT can enhance accuracy and consistency in DFIT analysis. It offers a path to integrate before- closure and after-closure analysis seamlessly, reduce the subjectivity in picking parameters like closure pressure, and improve field applicability (through faster and possibly automated analysis).

Data screening

Source data

The DFIT data sets for the 7 wells analyzed in-depth in the present study Table 1 were availed by three operators: Marathon (1 Bakken well, 2 Three Forks wells), Laredo Petroleum (2 Wolfcamp wells), and Hawkwood Energy (2 Eagle Ford wells). In addition to the DFIT data, the fracturing treatment files for these wells, as recorded during the actual stage-fracturing completion after conducting the DFIT, could also be consulted for additional verification of inferred ISIP and leak-off data. Consequently, our results were extensively cross-checked. In the further analysis of this paper, the study wells will simply be referred to with their numbers, followed by two capitals to denote the formation, as given in the first column of Table 1.

Table 1—Wells used in this study

Shale Play	Well	Shale Formation	Well Name	County	State	DFIT Start Date	Well Length (ft Kelly Bushing)	Permeability
1	BA-1	Bakken	LUCKY ONE 21-2H	Mountrail	North Dakota	15 May 2012	20,284	1 µD
2	TF-1	Three Forks	LUCKY ONE 21-2TFH	Mountrail	North Dakota	15 May 2012	20,247	
	TF-2	Three Forks	BEARS DEN 42-5TFH	Mountrail	North Dakota	30 Sep 2012	-	
3	WC-1	Wolfcamp	SUGG A 171 #650	Reagan	Texas	30 Sep 2015	-	0.1 µD
	WC-2	Wolfcamp	SUGG A 171 #850	Reagan	Texas	2 Oct 2015	-	
4	EF-1	Eagle Ford	HULLABALOO 2H	Brazos	Texas	2 Nov 2017	-	
	EF-2	Eagle Ford	HULLABALOO 3H	Brazos	Texas	2 Nov 2017	-	

Reconnecting recorded data to operational adjustments and reservoir response

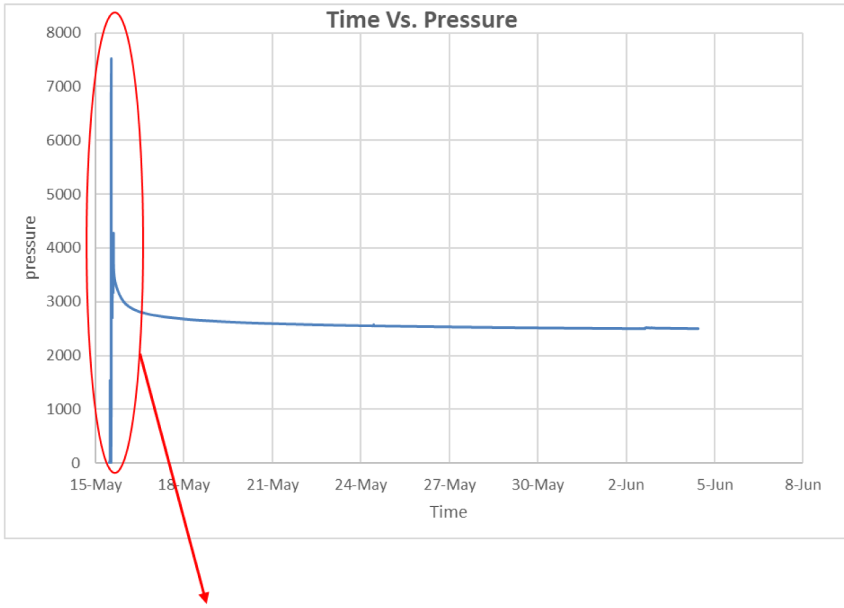
As a first step, we examined the primary pressure data logs in order to accurately link the recorded changes in pressure readings to distinctive events, such as (1) operational adjustments, and (2) actual reservoir responses. Based on the data inspection, cleaned-up graphs for each well were produced and commented on below, in order to identify which parts of the recorded curves are suitable for further DFIT-style analysis.

Bakken Well. The DFIT in Figure 1 shows the typical pressure response during a fracture injection test. Firstly, during the period from 12:45 PM to 1:45 PM, the well was first equalized with the equipment, recording an initial pressure of 500 psi. To ensure no gas was present in the upper section of the well, 200 psi were bled off. At 12:48 PM, the pump was activated, and pressure was recorded at 1-second intervals using the pump truck. The hydraulic fracture was opened after injecting 8.5 bbls at 7,550 psi with a rate of 1.3 bpm. Upon breakdown, the pressure dropped to 4,030 psi, after which the pumping rate was instantly increased to 2 bpm, marking the beginning of the Diagnostic Fracture Injection Test (DFIT). To ensure pressure stability, 2 bbls were initially injected, followed by an additional 15 bbls, resulting in an average injection pressure of 4,240 psi. The pump was then shut down instantly, and the Instantaneous Shut-In Pressure (ISIP) at 1 minute was recorded at 3,710 psi. Post shut-in, pressure monitoring continued for 30 minutes, with data recorded at 10-second intervals. The Shut-In Tubing Pressure (SITP) after 30 minutes was 3,370 psi, reflecting the initial pressure decline behavior during the fracture closure period.

Between 1:45 PM and 2:14 PM, the pump truck operator unintentionally began flowing back the well to initiate rig down (RD) operations without prior instructions. As a result, the well pressure was reduced to approximately 2,800 psi before the operator was instructed to stop. Following this event, the team consulted with Houston to assess the situation and determine the appropriate next steps

Between 2:15 PM and 3:15 PM, the equipment was retested to 7000 psi after connections were taken apart. At 2:25 PM, the well was equalized, and the pressure built up to 3160 psi. At 2:30 PM, the pump was brought online at 2 bpm. By 2:35 PM, after pumping 4 barrels, it was concluded that the pressure was stable Subsequently, 30 barrels were pumped at the same rate, as discussed with Houston engineering. The average pump pressure recorded was 4240 psi. At 2:55 PM, the ISIP at 1 minute was measured at 3680 psi, with a 17-second delay in fully shutting the rate down. For the next 30 minutes, pressure was recorded in 5-second intervals using the pump truck. By 3:15 PM, the SITP after 30 minutes was recorded at 3450 psi. Between 3:15 PM and 3:45 PM, the Rods Hot Oil systems were rigged down, and the valves leading to the Data Trap gauges remained open for continuous pressure gathering. From 3:45 PM to 06:00 AM, the well was secured using the Data Trap gauges and non-gauge wing valves, ensuring proper pressure monitoring and well integrity throughout the period.

(a)



(b)

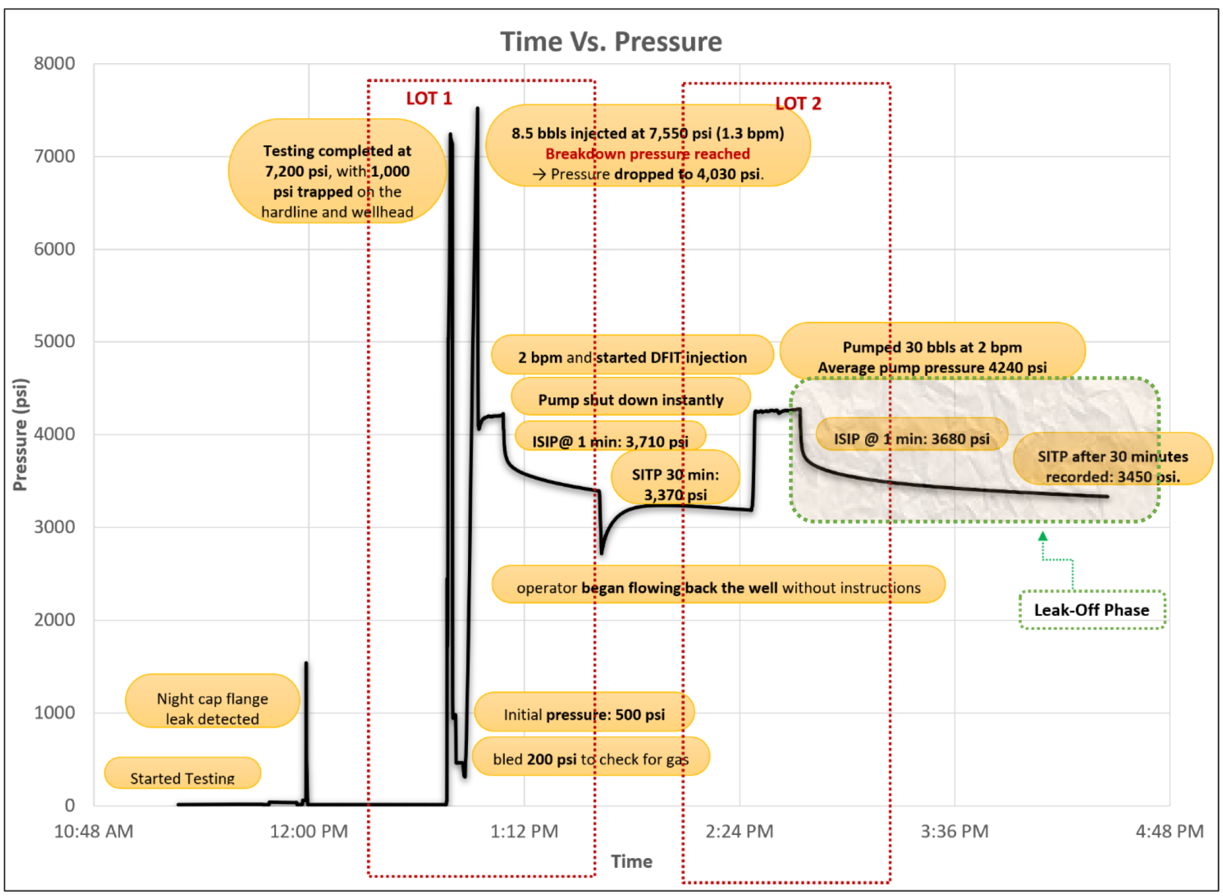


Figure 1—Pressure response during a fracture injection test in the Bakken Shale Formation, Mountrail County, North Dakota. (a) Full DFIT test data (20 day) since starting date 15 May 2012. (b) Critical steps during the initial DFIT operation, followed by the long leak-off phase that was studied in detail (see later in this paper).

Three Forks Well-1. Figure 2 presents the pressure profile obtained during a diagnostic fracture injection test (DFIT) performed in 2012 in the USA, operated by Marathon Oil Company.

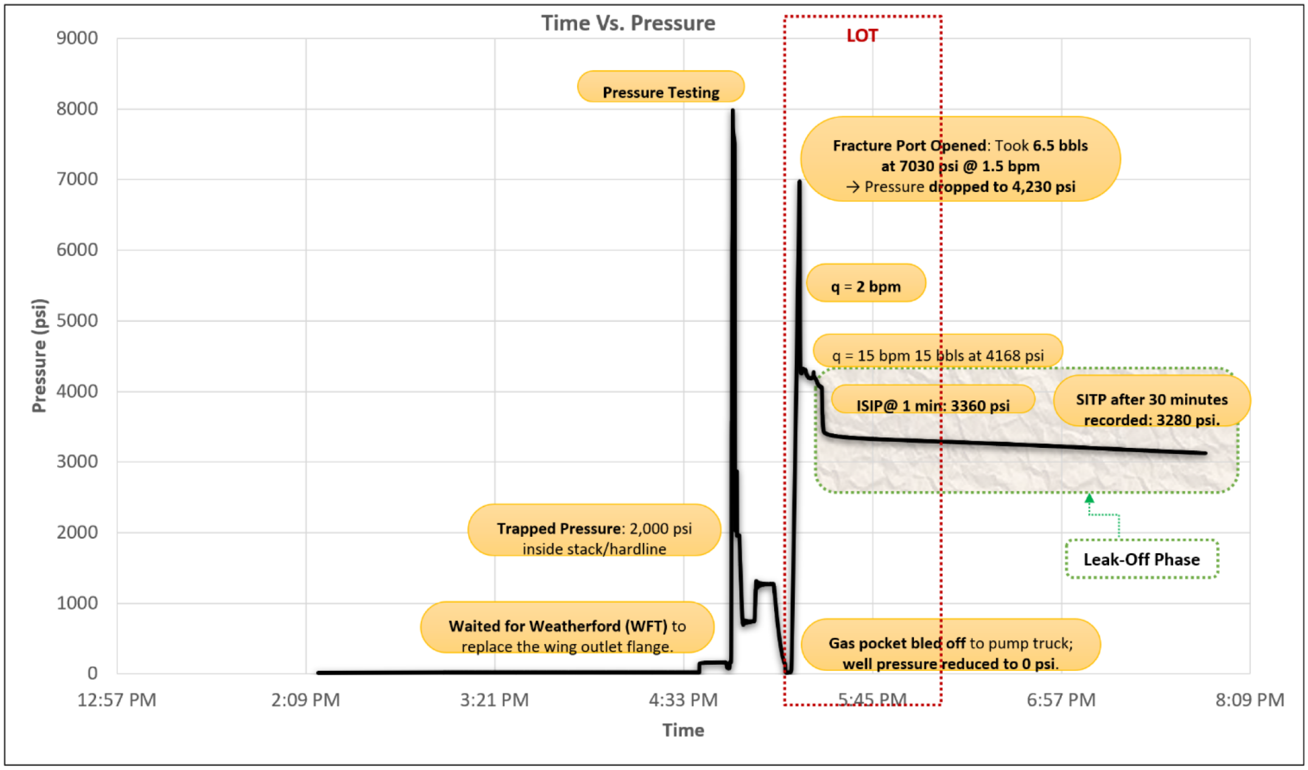


Figure 2—DFIT showing the typical pressure response during a fracture injection test in the Three Forks Shale Formation, Mountrail County, North Dakota. DFIT start date: 15 May 2012.

Initially, the pressure profile indicates a period of low pressure, followed by a sharp increase corresponding to the breakdown pressure necessary to open the hydraulic fracture. Following the peak pressure, a notable drop in pressure is observed, signifying fracture propagation and stabilization of fluid injection. The final stage of the profile illustrates the shut-in phase, characterized by a gradual pressure decline due to fluid dissipation into the formation.

Operationally, between 4:00 PM and 4:30 PM, activities were temporarily halted after discovering a seized swedge on the wing outlet. Operations resumed after Weatherford replaced the wing outlet flange. By 4:30 PM, the rods hot oil systems (RHOS) 15K hardline was rigged up (RU), and subsequent pressure tests on the hardline, frac stack, and Data Trap Gauge connections successfully reached 8,000 psi. Post- testing, a residual 2,000 psi was trapped inside the stack and hardline, prompting equalization of the well with surface equipment at 4:53 PM. Subsequently, to eliminate any residual gas pockets, the well pressure was reduced to 0 psi using the pump truck. At 5:15 PM, pumping operations commenced, recording pressures at 1-second intervals. Opening the hydraulic fracture required 6.5 barrels (bbls) of fluid injected at 7,030 psi with a flow rate of 1.5 barrels per minute (bpm). Immediately following fracture breakdown, the pressure sharply decreased to 4,230 psi, prompting an instantaneous increase in the pumping rate to 2 bpm, officially initiating the DFIT. Initial injection included 2 bbls of fluid to confirm stability, followed by an additional 15 bbls at an average sustained pressure of 4,168 psi. The Instantaneous Shut- In Pressure (ISIP) after one minute was recorded at 3,360 psi, accounting for a 17-second delay required to fully halt pumping. Subsequently, the pump truck continued pressure measurements every 5 seconds for 30 minutes, at which time the Shut-In Tubing Pressure (SITP) was measured at 3,280 psi. Between 6:15 PM and 6:45 PM, both the MBI and RHOS were rigged down (RD), while valves leading to the data trap gauges remained open for continued monitoring. From 6:45 PM to 6:00 AM, the well was secured using data trap gauges and non-gauge wing valves, ensuring pressure stability overnight.

Three Forks Well-2. Figure 3 shows the typical pressure response during a fracture injection test. During this Diagnostic Fracture Injection Test (DFIT), a sharp pressure increase was observed around 12:36 PM, reaching approximately 7,590 psi. This peak marks the breakdown pressure required to open the hydraulic fracture and initiate the fracturing process. Shortly after the peak, the pressure dropped to around 4,200 psi, indicating the start of fracture propagation as the fluid began entering the formation. This is a critical phase where the fracture extends due to fluid injection. Between 12:37 PM and 12:50 PM, the pumping rate was stabilized at 2 bpm, resulting in an average pressure of approximately 4,180 psi. This phase corresponds to the stabilized injection period of the DFIT, where pressure monitoring was conducted during fluid injection. At 12:50 PM, the pump was shut off, and the Instantaneous Shut-In Pressure (ISIP) was recorded at 4180 psi. This marks the transition to the shut-in phase, where pressure behavior after pump Stopping is analyzed.

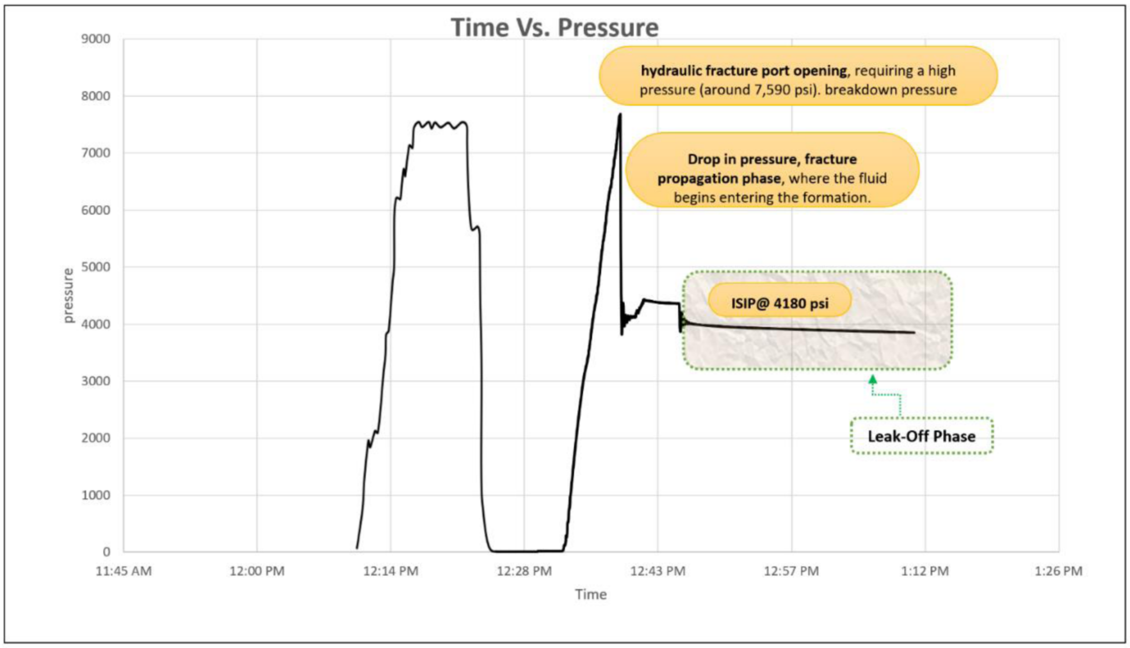


Figure 3—DFIT chart showing the typical pressure response during a fracture injection test in the Three Forks Shale Formation, Mountrail County, North Dakota. DFIT start date: 30 Sep 2012.

Wolfcamp Well-1. The Diagnostic Fracture Injection Test (DFIT) Figure 4 provides a detailed representation of the pressure and flow rate dynamics throughout the test. The chart captures four key phases: pressure buildup, fracture initiation, fracture propagation, and post-shut-in pressure decline. Initially, the pressure was equalized with the surface equipment, starting at 500 psi. The pressure steadily increased as the pump was activated, ultimately reaching the breakdown pressure (Pb) of approximately 8200 psi. This marked the opening of the hydraulic fracture. At this point, a sharp and immediate pressure drop to 3900 psi occurred, indicating the initiation of fracture propagation.

Following the breakdown, the pumping rate was increased to 7 bpm to stabilize the fluid injection process, maintaining an average pressure of approximately 3900 psi. This phase, labeled as the Fracture Propagation Phase (FPP), is characterized by stable pressure and flow rate conditions, reflecting effective fluid injection into the fracture. Once the desired fluid volume was injected, the pump was shut down, and the Instantaneous Shut-In Pressure (ISIP) was recorded at 3000 psi. This pressure represents the initial resistance of the fracture to closure when pumping stops.

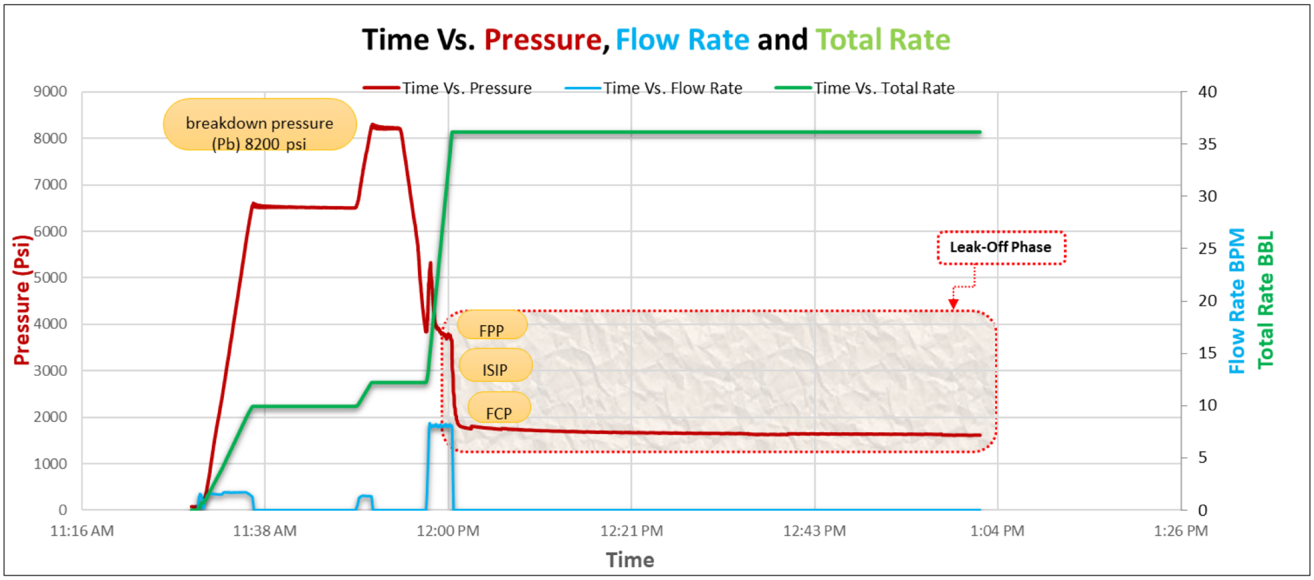


Figure 4—DFIT chart showing the typical pressure, flow rate and total volume response during a fracture injection test in Wolfcamp Shale Formation, Reagan County, Texas. DFIT start date: 30 Sep 2015.

During the post-shut-in phase, the pressure gradually declined as the fluid dissipated into the formation, a key indicator of fracture closure behavior. The Shut-In Tubing Pressure (SITP) after 30 minutes was recorded at 1800 psi, highlighting the pressure decay and providing valuable insights into the reservoir's permeability and fracture properties.

Wolfcamp Well-2. Figure 5 shows DFIT conducted on Wolfcamp Well-2 provides a detailed analysis of the pressure and flow rate dynamics throughout the injection process. The breakdown pressure was recorded at approximately 8,550. Then, the pressure showed a sharp decline to around 4,750 psi, indicating the fracture propagation phase, during which fluid began to enter the formation and extend the fracture. To sustain fluid injection and stabilize the propagation process, the flow rate was increased to 8 bpm, ensuring adequate fracture extension while maintaining an injection pressure of approximately 4,750 psi. A total of 30 barrels (BBL) of fluid was injected during this phase, contributing to the data necessary for assessing formation permeability and fracture closure characteristics. Once the target injection volume was reached, pumping was halted, and the Instantaneous Shut-In Pressure (ISIP) was recorded at about 2,900 psi. As expected, the pressure gradually declined over time, with a Shut-In Tubing Pressure (SITP) of 1,400 psi recorded after 30 minutes.

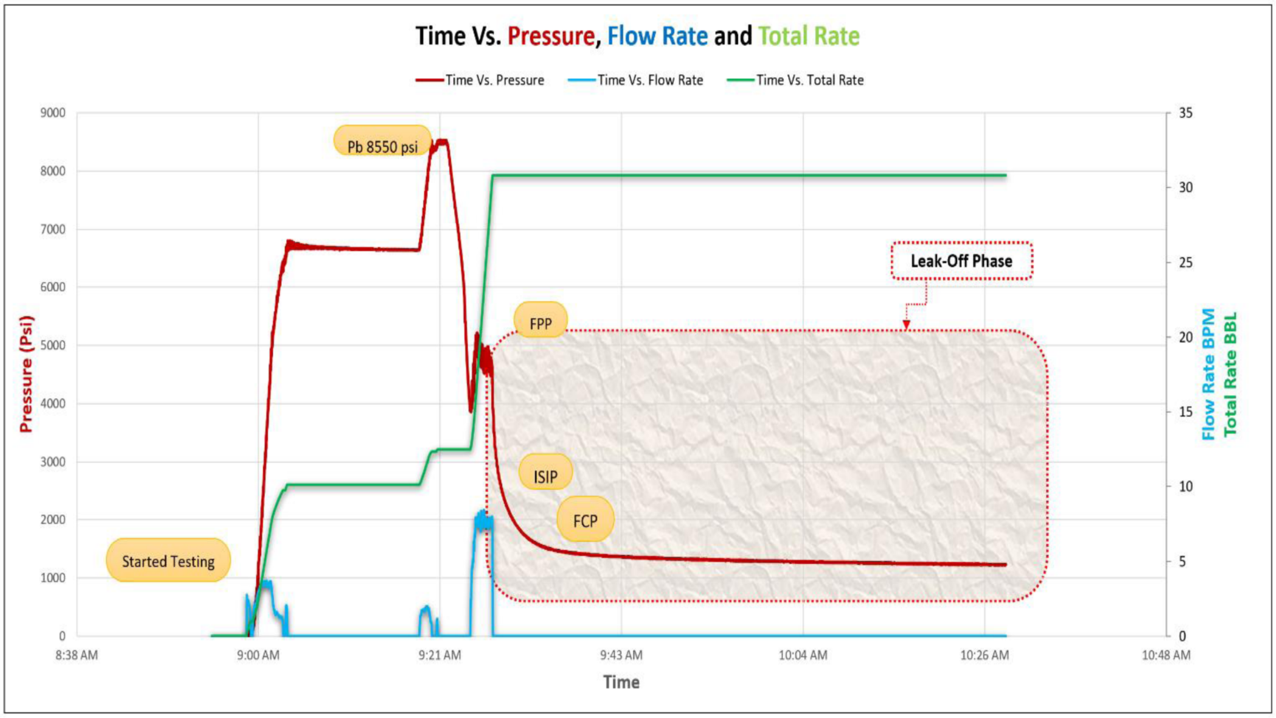
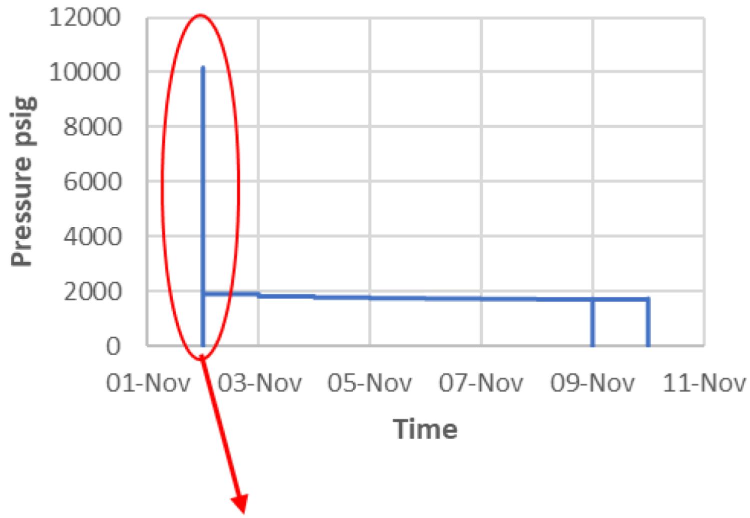


Figure 5—DFIT chart showing the typical pressure, flow rate and total volume response during a fracture injection test in Wolfcamp Shale Formation, Reagan County, Texas. DFIT start date: 2 Oct 2015.

Eagle Ford Well-1. The DFIT Figure 6 shows the typical pressure response during a fracture injection test. Initially, pressure increase sharply, reaching a breakdown pressure of about 10150 psi, where the formation fractures. After breakdown, the pressure drops to 4,850 psi where the fracture propagates. Pumping was stopped at about 9:35 AM for full leak-off. The well was then pressurized again to reach the Fracture Propagation Phase (FPP) at 9:43 AM. The Instantaneous Shut-In Pressure (ISIP) was recorded at 2,950 psi, followed by a gradual pressure decline during the fracture closure and fluid leak-off period, reaching a Shut-In Tubing Pressure (SITP) of 2,400 psi.

(a)



(b)

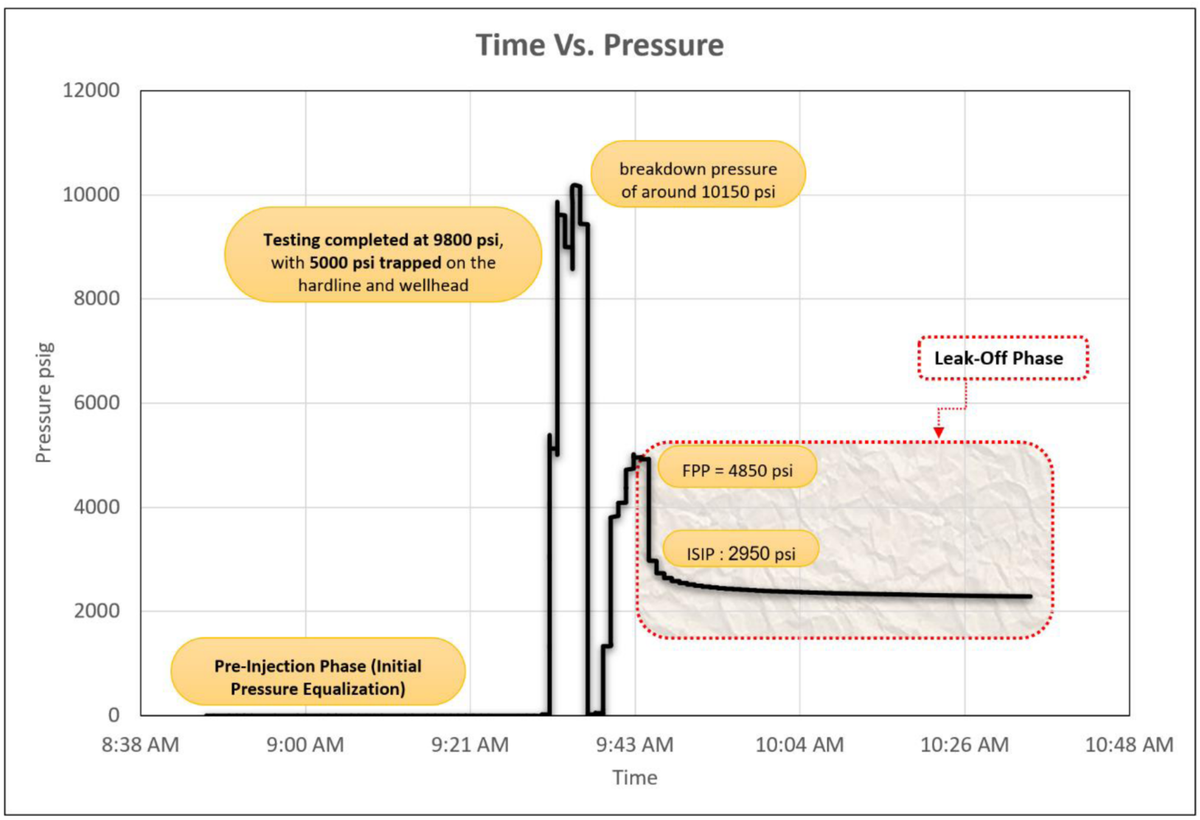


Figure 6—Pressure response during a fracture injection test in the Eagle Ford Shale Formation, Brazos County, Texas. (a) Full DFIT test data (8 days) for period 2-10 November 2017. (b) Critical steps during the initial DFIT operation, followed by the long leak-off phase that was studied in detail (see later in this paper).

Eagle Ford Well-2. Figure 7 illustrates the typical pressure response during a fracture injection test. Initially, pressure rises sharply, reaching a breakdown pressure of approximately 10000 psi, at which point the formation fractures. The well is then bled off to reduce pressure, after which the well is pressurized by increasing the injection rate until the fracture propagation phase is reached about 4,800 psi. The Instantaneous Shut- In Pressure (ISIP) is recorded at 3,300 psi, followed by a gradual pressure decline throughout the fracture closure and fluid leak-off period, reaching a Shut-In Tubing Pressure (SITP) of 2,500 psi.

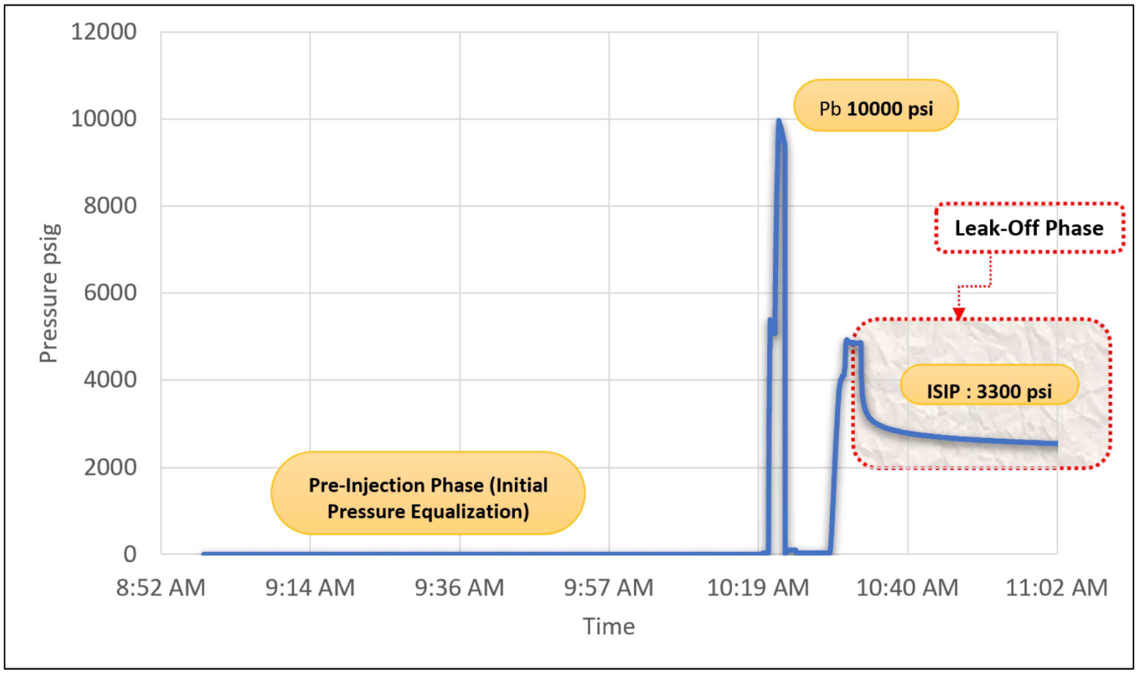


Figure 7—DFIT chart shows the typical pressure response during a fracture injection test in Eagle Ford Shale Formation, Brazos County, Texas. DFIT start date: 2 Nov 2017.

Summary data

Table 2 summarizes the key injection and pressure parameters recorded for the seven analyzed DFITs, organized by shale formation (Bakken, Three Forks, Wolfcamp, Eagle Ford). For each well, the breakdown pressure (required to initiate fracture), flow rates during injection, and the total fluid volumes pumped during different stages (breakdown, stabilization, DFIT injection, and second injection if applicable) are listed. The fracture propagation pressure reflects the estimated pressure required to keep the fracture open during propagation. The fracture propagation pressure represents the pressure required to keep the fracture open during propagation. Two parameters of particular diagnostic importance in DFIT analysis are the Instantaneous Shut-In Pressure (ISIP) and the Shut-In Tubing Pressure (SITP) measured 30 minutes after shut-in, which are commonly used to estimate leak-off behavior, reservoir permeability, and fracture closure. Note that for some wells (e.g., EF-1 and EF-2), operational data such as flow rate and total pumped volume were not reported in the field records and are therefore marked as unavailable.

Table 2—Key parameter and pump rates

Shale	Well	Breakdown Pressure (psi)	Flow Rate (bpm)	Total Fluid Pumped (BBL)	Fracture propagation pressure (psi)	Instantaneous Shut-In Pressure (ISIP) (psi)	Shut-In Tubing Pressure (SITP) (psi)
1	BA-1	7550	Initially 1.3 bpm, increased immediately after breakdown to 2 bpm	Total injected volume = 8.5 BBL (initial breakdown) + 2 BBL (initial stabilization) + 15 BBL (DFIT first injection + 30 BBL (second injection) = 55.5 BBL	4270	(1-minute): 3,710 psi	(after 30 minutes): 3,370 psi
2	TF-1	7000	Initially 1.5 bpm, then increased immediately to 2 bpmafter breakdown.	Total injected volume = 6.5 BBL (initial fracture opening + 2 BBL (initial stabilization) + 15 BBL (DFIT injection) = 23.5 BBL	4230	3,360	3,280 psi after 30 minutes shut-in.
	TF-2	7,590	2 bpm	27.5 BBL *(standard assumption)*	4200	4180 psi (corrected)	3900
3	WC-1	8,200	7 bpm	35 BBL	3900	3,000	1,800
	WC-2	8550	8 bpm	30 BBL	4750	2900	1400
4	EF-1	10150	Not provided	33 BBL	4850	2950	2400
	EF-2	10000	Not provided	31 BBL	4800	3300	2500

For each of the seven study wells, the post-fracture propagation pressure fall-off data was isolated, and the plots were compiled with timescale in seconds, visualizing pressure falloff after pumps were shut-in Figure 8

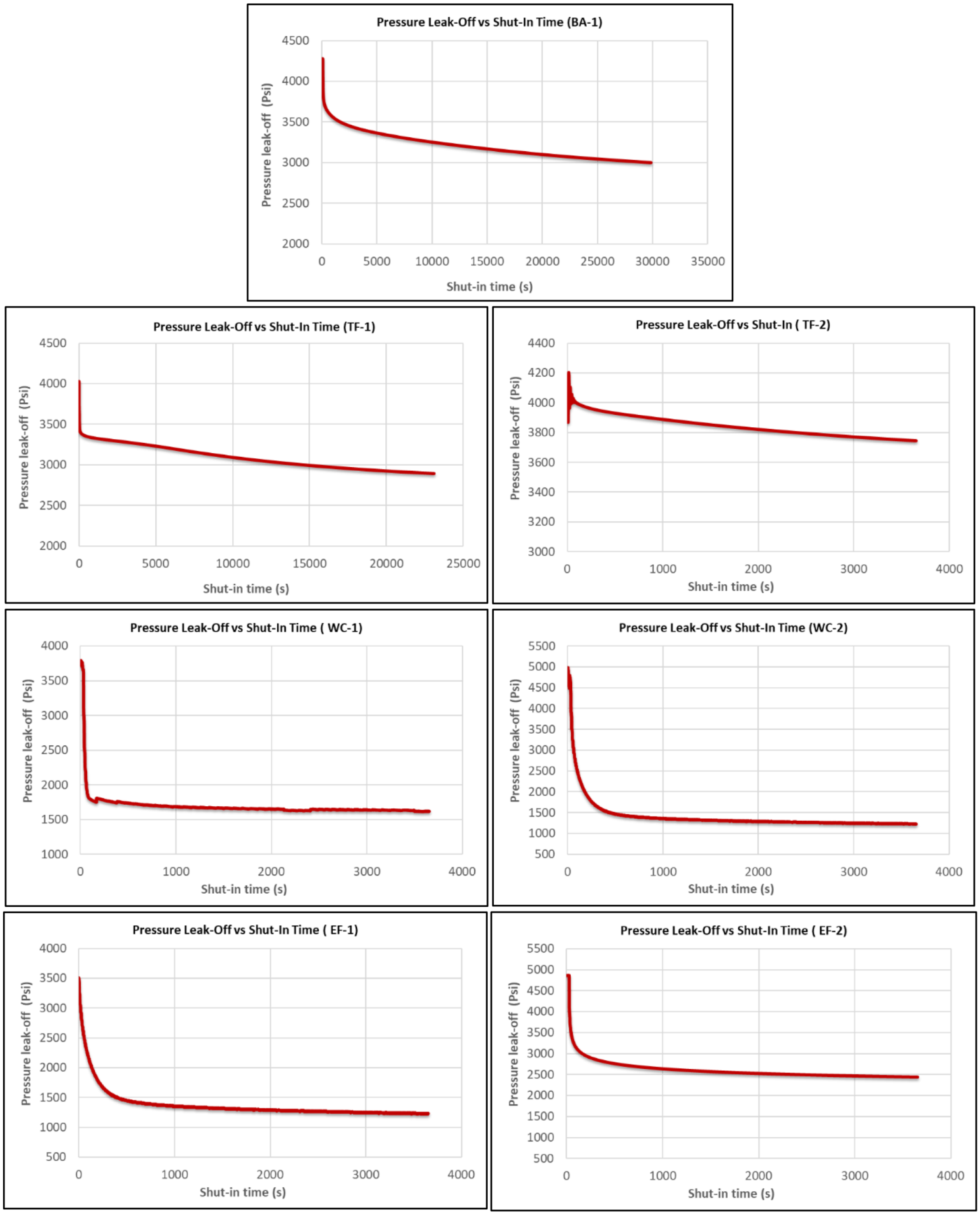


Figure 8—Leak-off curves for the seven wells, using well identification numbers given in Table 1.

Fracture growth and pressure transient analysis

This section explained the theoretical framework developed to provide a concise and accurate new methodology to match DFIT data, using logical steps. The overall workflow is presented in Figure 9 and further summarized in Appendix A.

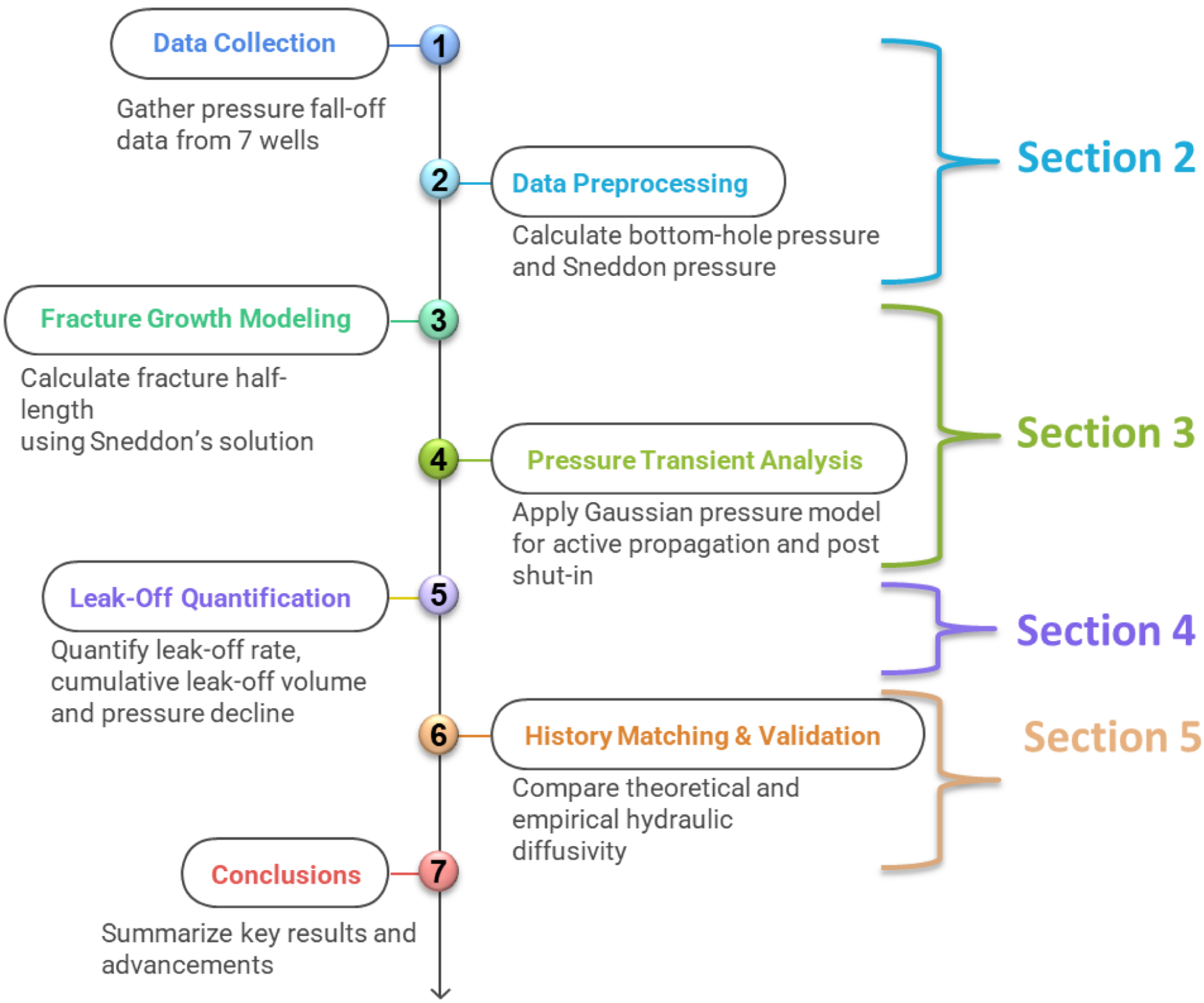


Figure 9—Workflow diagram of the Gaussian Pressure Transient method for DFIT analysis, showing Steps 1-7 from data collection to conclusion.

Fracture growth modeling

Determining the dimensions of the DFIT mini-frac. For opening of the mini-frac, Sneddon's solution is used (Sneddon, 1946; Sneddon and Elliott, 1946), which gives for each net pressure, the relationship between the two half-lengths of the fracture Figure 10, as follows (Weijermars and Oshaish, 2025):

$$P_S = \frac{E}{2(1-\nu^2)} \frac{\alpha_f}{y_f} \tag{1}$$

where:

- P_S : Sneddon pressure psi
- E : Young's modulus psi
- ν : Poisson's ratio
- α_f : fracture half-width ft
- y_f : fracture half-length ft

The solution assumes a plane strain boundary condition for a Griffith crack in an infinite elastic continuum opening due to an internal net pressure (Sneddon pressure), P_S.

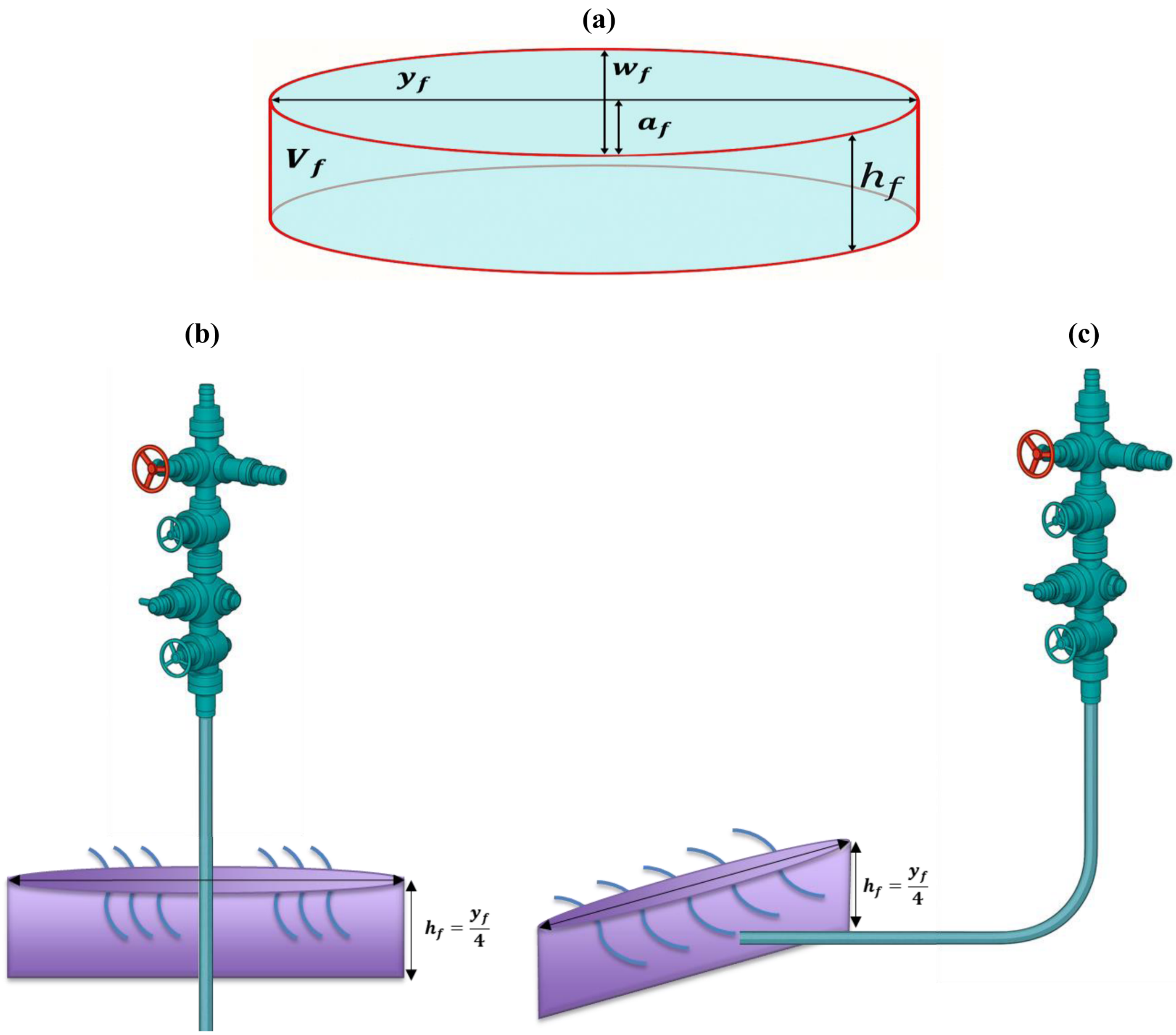


Figure 10—Mini-frac dimensions and notation used: (a) elliptical fracture dimensions; (b) vertical fracture; (c) Horizontal fracture. The analysis of the fracture growth is not affected by the well orientation.

Using full fracture-aperture (or fracture-width, w_f, such that $\alpha_f = w_f/2$), Eq. (1) becomes (Weijermars and Oshaish, 2025):

$$P_S = \frac{E}{4(1-\nu^2)} \frac{w_f}{y_f} \tag{2}$$

Given certain Young's Modulus, E, and Poisson ratio, ν, the ratio w_f/y_f can be computed. However, for the leak-off computation, we need to solve for the final y_f achieved at the end of the mini-frac test. For that purpose, an additional volume balance equation can be used, using the total fluid volume, V_f, stored in the elliptical fracture at the time of well shut-in after the fracture propagation phase (Weijermars and Oshaish, 2025):

$$V_f = \frac{\pi}{2} h_f y_f \alpha_f = \frac{\pi}{4} h_f y_f w_f \tag{3}$$

where:

- V_f : total fracture volume ft³
- h_f : fracture height ft
- y_f : fracture half-length ft
- w_f : fracture width ft

Geometric assumption to simplify the model. For practical purposes we assume the mini-frac height, h_f, stays limited and at all times grows at a rate half that of the fracture propagation rate, such that the total fracture length/ fracture height ratio stays 4 at all times: $y_f/h_f = 4$. Substituting this assumption into Eq. (3) gives:

$$V_f = \frac{\pi}{4} \cdot \frac{y_f}{4} \cdot y_f \cdot w_f = \frac{\pi}{16} \cdot y_f^2 \cdot w_f \tag{4}$$

Rewriting Eq. (4) in terms of fracture width w_f:

$$w_f = \frac{16 V_f}{\pi y_f^2} \tag{5}$$

Substituting Eq. (5) into Eq. (2) for the Sneddon pressure:

$$P_S = \frac{E}{4(1-\nu^2)} \frac{16 V_f}{\pi y_f^3} = \frac{4E}{\pi(1-\nu^2)} \frac{V_f}{y_f^3} \tag{6}$$

Rearranging Eq. (6) to solve for the fracture half-length, y_f, gives

$$y_f = \left(\frac{4E}{\pi(1-\nu^2)} \frac{V_f}{P_S} \right)^{1/3} \tag{7}$$

This is the final expression used to compute the fracture half-length.

Inputs for determining fracture half-length of the DFIT mini-frac. For each reservoir and each well studied, the Sneddon pressure will be different, because it is determined by:

$$P_S = P_F - \sigma_3 \tag{8}$$

The value of P_F is the pressure recorded directly at the end of the mini-frac, and the Sneddon pressure is a net pressure obtained by subtracting the minimum tectonic stress σ_3. Eq. (8) assumes the mini-frac opens perpendicular to the least principal stress, which will be the case at all times based on fundamental geotechnical principles.

The inputs required for solving the final half-length, y_f, with Eq. (7) for the mini-frac are: P_F (from well test), σ_3 (from breakout test and well test), V_f (from DFIT pump log), E and ν (dynamic values from sonic log and static values from geomechanical tests on cores). These parameters are compiled from our available sources in Table 3, which allowed us to solve for y_f using Eq. (7).

Table 3—Key parameter for final half-length calculation

Well	$P_{F@Surface}$ (psi)	$P_{Hydrostatic}$ (psi)	$P_{F@Downhole}$ (psi)	σ_3 (psi)	$P_S = P_{F@Downhole} - \sigma_3$ (psi)	E (psi) (5 GPa)	v	V_f (bbl)	V_f (ft^3) 1 bbl=5.615 ft^3	y_f (ft)
BA-1	4267	4336	8603	500	8103	725188	0.25	55.5	311.63	33.6
TF-1	4029	4336	8365	500	7865	725188	0.25	23.5	131.95	25.5
TF-2	4085	4336	8421	500	7921	725188	0.25	27.5	154.41	26.8
WC-1	3731	4336	8067	500	7567	725188	0.25	35	196.52	29.5
WC-2	4976	4336	9312	500	8812	725188	0.25	30	168.45	26.6
EF-1	3507	4336	7843	500	7343	725188	0.25	33	185.29	29.2
EF-2	4854	4336	9190	500	8690	725188	0.25	31	174.06	27.0

In mini-frac tests, the Instantaneous Shut-In Pressure (ISIP) and the fracture propagation pressure are typically measured at the surface (denoted as $P_{F@Surface}$). However, the actual fracture propagation pressure at the reservoir depth is affected by the hydrostatic head of the fluid column in the wellbore. Therefore, to determine the bottom-hole pressure ($P_{F@downhole}$), the hydrostatic pressure must be added to the surface-measured pressure.

The bottom-hole pressure is computed as:

$$P_{F@downhole} = P_{F@Surface} + P_{hydrostatic} \tag{9}$$

Using the standard hydrostatic gradient conversion, the equation becomes:

$$P_{F@downhole} = P_{F@Surface} + 0.052 \times \rho \times h \tag{10}$$

where:

- $P_{F@downhole}$ = Bottom-hole pressure (psi)
- $P_{F@Surface}$ = Surface-measured pressure (psi)
- $P_{hydrostatic}$ = Hydrostatic pressure from the fluid column
- ρ = Fracturing fluid density (ppg)
- h = Well depth (ft)

For example, with slickwater (ρ = 8.34 ppg) and a well depth of 10,000 ft**,** the hydrostatic pressure would be: $P_{hydrostatic} = 0.052 * 8.34 * 10{,}000 = 4336\, psi$. Thus, the bottom-hole pressure can be accurately estimated and used in subsequent fracture modeling and pressure balance calculations.

Table 3 summarizes the sequential steps used to compute the fracture half-length y_f for each well. The procedure begins with the surface-measured fracture pressure $P_{F@Surface}$, which is corrected for hydrostatic pressure to obtain the bottom-hole pressure $P_{F@downhole}$. The Sneddon pressure Ps is then calculated by subtracting the minimum in-situ stress σ_3. Along with elastic rock properties (Young's modulus E and Poisson's ratio v) and the total injected fluid volume V_f, the final fracture half-length is determined using Eq. (7). The summary of Table 3 allows for a direct comparison across wells and provides insight into how input parameters influence fracture geometry. The fracture half-length of the mini-frac test ranges between 26 to 34(ft). This is significantly shorter than the typical fracture half- length of a full hydraulic fracturing treatment where the fracture half-length will range between 110 and 240 (Weijermars & Oshaish, 2025). The difference arises due to the limited volume pumped in the mini- frac test.

Pressure Transient Analysis

Determining the pressure decline near the fracture. To determine the pressure change related to fracturing treatment interventions, we employ the Gaussian pressure transient solution (Weijermars, 2022). The pressure at a given location and time, $P_{(x,t)}$, in the reservoir section normal to the created mini-frac will experience a pressure-transient advance and local increase in reservoir pressure due to fluid injection. The new reservoir pressure, $P_{(x,t)}$, is defined by the initial reservoir pressure, P_0, plus the imposed pressure change, $\Delta P_{(x,t)}$:

$$P_{(x,t)} = P_0 + \Delta P_{(x,t)} \tag{11}$$

The scenario uses a physics-based solution from Tian and Weijermars (2025); the pressure change is given by:

$$\Delta P_{(x,t)} = \Delta P_0 \left[1 - erf\left(\frac{x}{2\sqrt{D_h t}} \right) \right] \tag{12}$$

So, the actual pressure in the reservoir space adjacent to the hydraulic fracture can be computed from

$$P_{(x,t)} = P_0 + \Delta P_{(x,t)} = P_0 + \Delta P_0 \left[1 - erf\left(\frac{x}{2\sqrt{D_h t}} \right) \right] \tag{13}$$

Table 4 summarizes the key input parameters assumed in the pressure modeling of the reservoir system. The fracture pressure (P_f) and initial reservoir pressure (P_0) were used to calculate the pressure differential (ΔP_0). The hydraulic diffusivity (D_h) governs the rate of pressure propagation through the formation and is expressed in ft²/minute. The values adopted for porosity ϕ, viscosity μ, and compressibilities (c_r, c_f, c_t) were sourced from Weijermars (2022), who applied these parameters in the context of Gaussian pressure transient modeling for multi-fractured wells in the. These values form the basis for the subsequent analytical and numerical evaluation of leak-off and pressure distribution behavior.

Table 4—Input parameters used in pressure diffusion calculation

Assumptions		
variable	**value**	**unit**
k	1.05E-18	ft^2
ϕ	0.06	-
μ	1.45038E-07	psi.s
c_r	6.89476E-08	psi^{-1}
c_f	6.89476E-07	psi^{-1}
c_t	1.06179E-07	psi^{-1}
D_h	**0.001136364**	**ft^2/s**
	0.06818184	**ft^2/minute**

Table 5 shows the pressure values (psi) computed at various distances (ft) from the fracture face, using Eq. (13), for a shale reservoir. The results are shown for time intervals ranging from 1 minute to 8.8 minutes, with 0.5 ft spacing in radius. The values illustrate how pressure changes dissipate spatially into the formation over time, providing insights into the reservoir's pressure front propagation and supporting the analysis of fluid leak-off and formation diffusivity.

Table 5—Pressure (psi) vs radius (ft) at different time intervals (min)

Time (min) \ Radius (ft)	0.0	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0	3.3	3.6	3.9	4.2
1.0	3507	2179	1468	1265	1234	1231	1231	1231	1231	1231	1231	1231	1231	1231	1231
1.3	3507	2315	1582	1305	1241	1232	1231	1231	1231	1231	1231	1231	1231	1231	1231
1.6	3507	2416	1684	1354	1254	1234	1231	1231	1231	1231	1231	1231	1231	1231	1231
1.9	3507	2496	1774	1406	1273	1238	1232	1231	1231	1231	1231	1231	1231	1231	1231
2.2	3507	2560	1853	1459	1296	1245	1233	1231	1231	1231	1231	1231	1231	1231	1231
2.5	3507	2613	1923	1511	1322	1254	1236	1232	1231	1231	1231	1231	1231	1231	1231
2.8	3507	2659	1986	1562	1350	1266	1239	1233	1231	1231	1231	1231	1231	1231	1231
3.1	3507	2698	2041	1609	1379	1279	1244	1234	1232	1231	1231	1231	1231	1231	1231
3.4	3507	2732	2092	1655	1409	1294	1250	1236	1232	1231	1231	1231	1231	1231	1231
3.7	3507	2762	2137	1698	1438	1310	1257	1238	1233	1231	1231	1231	1231	1231	1231
4.0	3507	2789	2179	1739	1468	1327	1265	1241	1234	1232	1231	1231	1231	1231	1231
4.3	3507	2813	2217	1777	1498	1345	1274	1245	1235	1232	1231	1231	1231	1231	1231
4.6	3507	2835	2252	1813	1526	1364	1283	1249	1237	1233	1231	1231	1231	1231	1231
4.9	3507	2855	2285	1848	1554	1382	1294	1254	1239	1233	1232	1231	1231	1231	1231
5.2	3507	2873	2315	1880	1582	1401	1305	1260	1241	1234	1232	1231	1231	1231	1231
5.5	3507	2890	2343	1911	1609	1421	1317	1266	1244	1235	1232	1231	1231	1231	1231
5.8	3507	2906	2369	1940	1634	1440	1329	1272	1247	1236	1233	1232	1231	1231	1231
6.1	3507	2920	2393	1968	1659	1459	1341	1280	1250	1238	1233	1232	1231	1231	1231
6.4	3507	2934	2416	1994	1684	1478	1354	1287	1254	1240	1234	1232	1231	1231	1231
6.7	3507	2946	2438	2019	1707	1496	1367	1295	1258	1242	1235	1232	1231	1231	1231
7.0	3507	2958	2458	2043	1730	1515	1380	1303	1263	1244	1236	1233	1232	1231	1231
7.3	3507	2969	2477	2066	1752	1533	1393	1311	1268	1247	1237	1233	1232	1231	1231
7.6	3507	2979	2496	2088	1774	1551	1406	1320	1273	1249	1238	1234	1232	1231	1231
7.9	3507	2989	2513	2109	1795	1569	1420	1329	1278	1252	1240	1234	1232	1231	1231
8.2	3507	2999	2529	2129	1815	1586	1433	1338	1284	1255	1241	1235	1233	1232	1231
8.5	3507	3007	2545	2149	1834	1603	1446	1347	1290	1259	1243	1236	1233	1232	1231
8.8	3507	3016	2560	2167	1853	1620	1459	1357	1296	1262	1245	1237	1233	1232	1231

To evaluate the spatial propagation of the pressure front, we applied Eq. (13) to simulate how pressure dissipates away from the fracture face over time. Figure 11 shows the pressure profiles along the reservoir distance (in ft) at successive 0.3-minute intervals, starting from t = 1 minute up to t = 10 minutes, for the Eagle Ford well.

Each curve in Figure 11 represents the pressure gradient from the fracture face outward into the reservoir at a given time. As time progresses, the pressure front advances further, and the pressure gradient becomes more distributed. This profile provides direct insight into the pressure diffusion mechanism and reservoir response, forming the foundation for estimating leak-off coefficients and formation permeability in the next stages of the analysis.

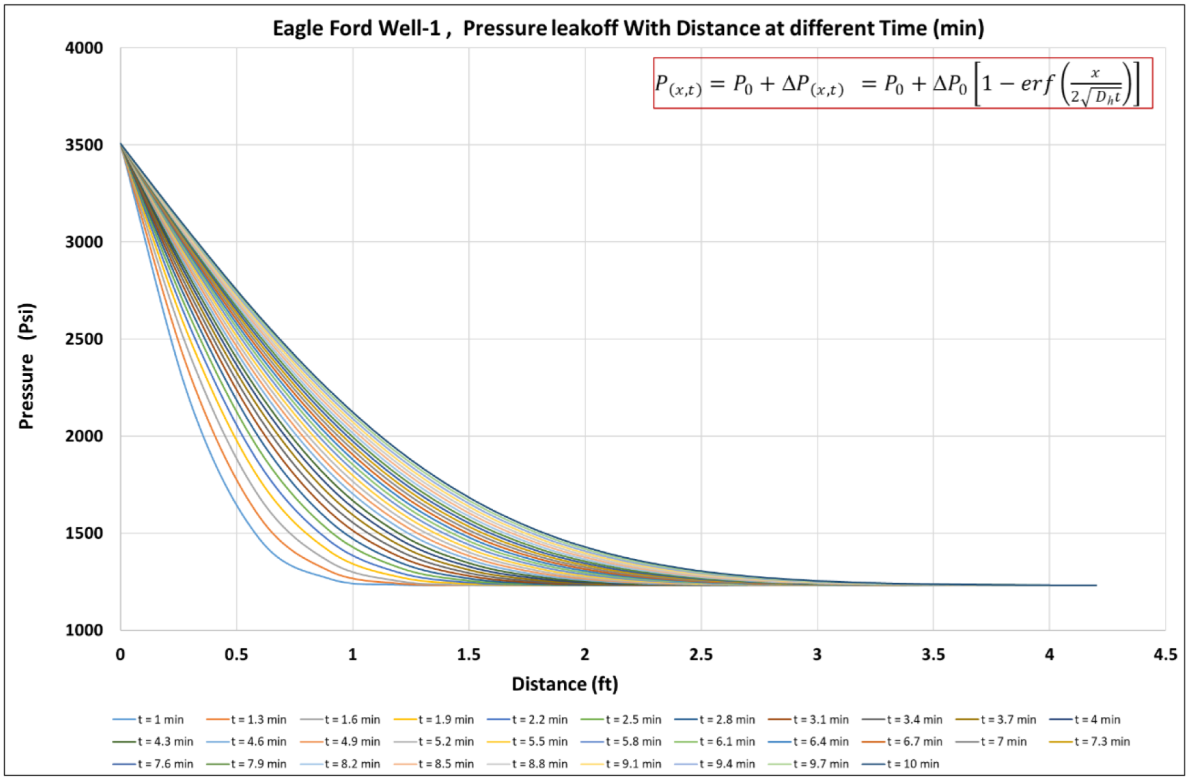


Figure 11—Pressure leak-off profiles vs. distance at different times for the Eagle Ford well.

Leak-off quantification

Analytical workflow for leak-off quantification in DFIT

The workflow in Figure 12 summarizes the stepwise procedure developed for quantifying leak-off during DFIT analysis. The process begins with the determination of the instantaneous leak-off rate $q_L(t)$ based on fracture and reservoir parameters. The leak-off rates are then accumulated over time to obtain the cumulative leak-off volume $V_L(t)$. Finally, this cumulative volume is coupled with compressibility and volume balance relationships to compute the pressure decline in the fracture due to leak-off $P_{fall\text{-}off}(t)$. This workflow establishes a consistent analytical pathway that links the leak-off rate, leak-off volume, and fracture pressure decline, and it provides the foundation for the newly derived analysis method DFIT.

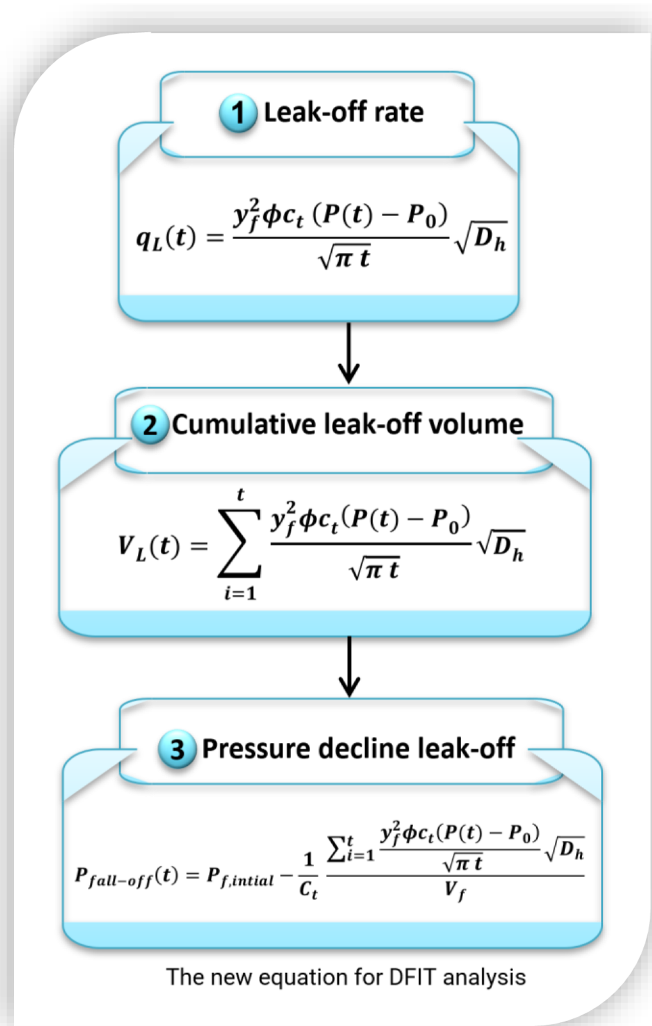


Figure 12—Analytical workflow for DFIT leak-off quantification linking (1) leak-off rate, (2) cumulative volume, and (3) pressure decline.

Determining the leak-off rate for a fracture

The pressure gradient $\nabla P(x, t)$ near the hydraulic fracture follows from:

$$\nabla P(x, t) = -\frac{d\,P_{(x,t)}}{dx} = -\frac{d\left(P_0 + \Delta P_0\left(1 - erf\left(\frac{x}{2\sqrt{D_h t}}\right)\right)\right)}{dx} \tag{14}$$

The application of Eq. (14) to Eq. (12) gives the pressure gradient solution:

$$\nabla P(x, t) = -\frac{\Delta P_0}{\sqrt{\pi D_h t}} . e^{-\frac{x^2}{4 D_h t}} \tag{15}$$

On the fracture surface itself, located at x=0, the pressure gradient simplifies to:

$$\nabla P(t) = -\frac{\Delta P_0}{\sqrt{\pi D_h t}} \tag{16}$$

The transient leak-off rate, $q_L(t)$, from the mini-frac can be computed from Darcy's Law:

$$q_L(t) = -2A\frac{k}{\mu}\nabla P(t) \tag{17}$$

A is the total surface area of the fracture; factor 2 in Eq. (17) occurs because the leak-off occurs into the rock matrix flanking the fracture at either side. In Eq. (17), also appears permeability, k, and viscosity of the injection fluid, μ. The resulting leak-off function is given by substitution of Eq. (16) into Eq. (17):

$$q_L(t) = -\frac{2Ak}{\mu}\frac{\Delta P_0}{\sqrt{\pi D_h t}} \tag{18}$$

The fracture surface area equal to $A = 2\, y_f h_f$ (twice the fracture half-length must be used). We assumed in Eq. (4) that $y_f/h_f = 4$ which gives $A = \frac{y_f^2}{2}$. Insertion into Eq. (18) gives:

$$q_L(t) = -\frac{y_f^2 k}{\mu}\frac{\Delta P_0}{\sqrt{\pi D_h t}} \tag{19}$$

Next, we simplify and remove k by using the hydraulic diffusivity expression with primary values. The hydraulic diffusivity contains the classic reservoir parameters, as follows:

$$D_h = \frac{k}{\phi \mu c_t} \tag{20}$$

with permeability, k, porosity, ϕ, fluid viscosity, μ, and total compressibility c_t given by $c_t = (1-\phi)c_{rock} + \phi c_{fluid}$, , and D_h was computed using Eq. (20) based on the input classic reservoir parameters summarized in Table 4.

Insertion of Eq. (20) in Eq. (19) gives:

$$q_L(t) = -\frac{y_f^2 \phi c_t \Delta P_0}{\sqrt{\pi t}}\sqrt{D_h} \tag{21}$$

In injection scenarios (like in DFITs), especially during the leak-off phase, we often report the leak-off rate $q_L(t)$ as a positive value, even though Darcy's law gives a negative sign (indicating direction of flow).

Table 6—Input parameters used in leak-off rate calculation EF-1, using Eq. (22)

Parameter	Symbol	Value	Unit
Fracture half-height	y_f	29.2	ft
Porosity	ϕ	0.06	-
Total compressibility	c_t	1.06179E-07	psi^{-1}
fracture pressure at shut-in time	P_f	3507	psi
Reservoir Pressure	P_0	1231	psi
Pressure difference	$\Delta P(t) = P(t) - P_0$	P(t) - 1231	psi
Fracture volume	V_f	185.295	ft^3
Computed Hydraulic diffusivity	D_h	**0.0011**	ft^2/s
History Matched Hydraulic diffusivity	D_h	**0.093**	ft^2/s

Here ΔP was treated the pressure difference as time-dependent ($\Delta P(t) = P(t) - P_0$). Therefore, applying this in Eq. (21) will get,

$$q_L(t) = -\frac{y_f^2 \phi c_t (\boldsymbol{P(t)} - P_0)}{\sqrt{\pi t}}\sqrt{D_h} \tag{22}$$

The leak-off rate was derived starting from the pressure gradient near the fracture surface, as given by Eq. (14–16), and then applying Darcy's Law (Eq. 17). By substituting the fracture surface area and expressing permeability in terms of hydraulic diffusivity, the leak-off rate formulation was simplified to Eq. (21). To account for the time-dependent nature of the pressure decline in the hydraulic fracture, the pressure difference was expressed as $\Delta P(t)=P(t)-P_0$ see Figure 13, resulting in the final expression for the transient leak-off rate given in Eq. (22). It should be noted that $\Delta P(t)$ represents a time-dependent change of the pressure difference in the hydraulic mini-fracture as determined by the difference of the pressure in the fracture $P(t)$ and the original reservoir pressure P_0 (Figure 13). While delta $\Delta P_{(x,t)}$ of Eqs. (11)-(13) gave the pressure change in the reservoir adjacent to the fracture in the reservoir space, Eq. (22) uses $\Delta P(t)$ for the pressure change in the fracture at x=0.

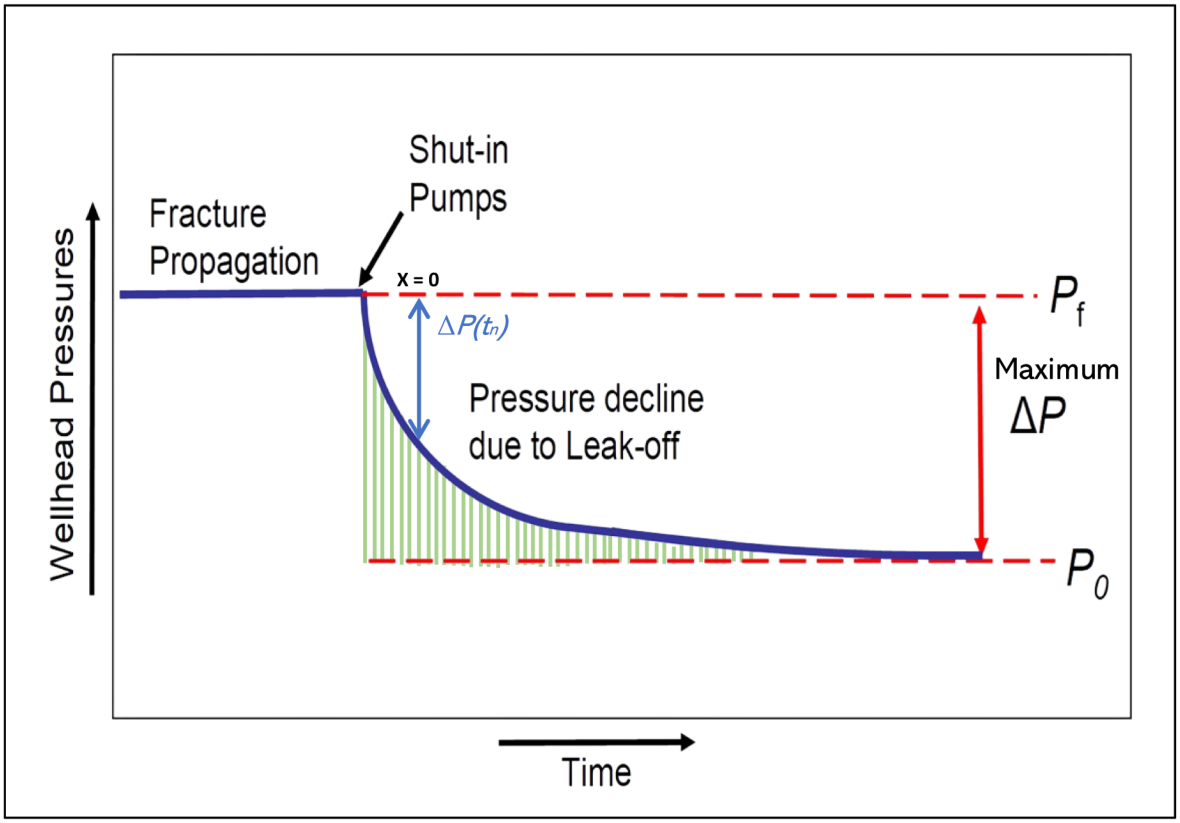


Figure 13—Determination of Δ*P* from the wellhead pressure readings in the pressure fall-off test.

Since P_F is routinely given as a wellhead pressure reading (EPA, 1994), and the reservoir pressure P_0 is not always known initially, for determination of maximum ΔP, we use the wellhead pressures for P_F and P_0 as recorded during the leak-off test Figure 13.

Thus, the maximum ΔP was determined for each well as given in Table 7. These values are useful only for the constant ΔP assumption, which can serve as a baseline approximation for very early shut-in conditions when the fracture pressure has not yet declined significantly. However, for the main workflow of this study, where the fracture pressure decreases continuously after shut-in, we employed the time- dependent pressure difference $\Delta P(t)$.

Table 7—Pressure differential calculation

Shale Play	Well	FPP (psi)	Final Pressure after Leak-off (psi)	ΔP (psi)
1	BA-1	4270	3300	935
2	TF-1	4230	3100	1130
	TF-2	4200	3800	400
3	WC-1	3900	1450	2450
	WC-2	4750	1200	3550
4	EF-1	4850	1220	3630
	EF-2	4800	2450	2350

Figure 14 presents the leak-off rate versus shut-in time for the Eagle Ford DFIT example, using fracture and reservoir parameters listed in Tables 3 and 6. The results indicate a sharp decline in the leak-off rate immediately after shut-in, followed by stabilization at a very low value. This confirms that the leak-off rate is negligibly small on the time scale of the fracturing treatment, consistent with the behavior expected in ultra-low-permeability shale reservoirs.

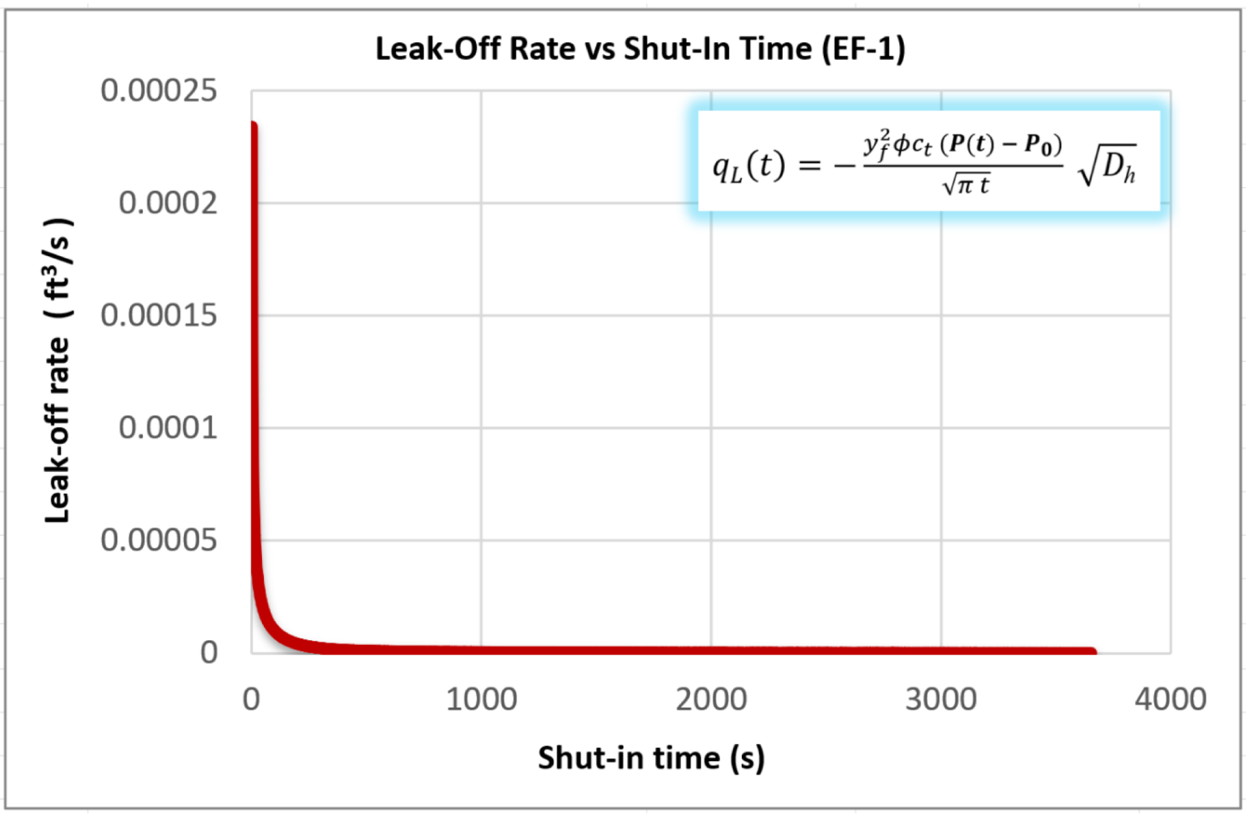


Figure 14—leak-off rate, $q_L(t)$, from fracture slot into shale. Fluid rate is in ft³ per *minute*. Used are k =100 nDarcy, μ =1 cPoise, P_0=1231 psi, and $P(t)$ in psi (empirical input from fall-off curve)

Determining the cumulative leak-off volume for a fracture

The cumulative leak-off volume at a given time can be obtained directly from the leak-off rate by numerical summation. Because the fracture pressure declines continuously after shut-in, the pressure difference $\Delta P(t)$ is inherently time dependent.

Analytical integration attempts were also tested, but these did not provide physically reliable results for post-shut-in conditions. The full derivation and discussion of the analytical solution are included in Appendix B for reference.

To capture the true transient behavior, we adopt a stepwise numerical summation approach:

$$V_L(t) = Q_L(t) = \sum_{i=1}^{t} q_{l,i} \tag{23}$$

By inserting Eq. (22) in Eq. (23) we will get:

$$V_L(t) = Q_L(t) = \sum_{i=1}^{t} \frac{y_f^2 \phi c_t (\boldsymbol{P(t)} - \boldsymbol{P_0})}{\sqrt{\pi t}} \sqrt{D_h} \tag{24}$$

This approach accumulates the leak-off volume incrementally over time steps and properly accounts for the time-varying pressure difference $\Delta P(t)$. It therefore provides an accurate and physically consistent estimate of the cumulative leak-off volume over the shut-in period. Figure 15 illustrates the cumulative leak-off volume versus shut-in time for well EF-1, as computed using the numerical summation method of Eq. (24). The curve shows the expected trend where the cumulative volume increases rapidly at early shut-in times before gradually approaching stabilization. This behavior highlights the ability of the proposed approach to consistently represent the time-dependent nature of fluid leak-off in unconventional reservoirs.

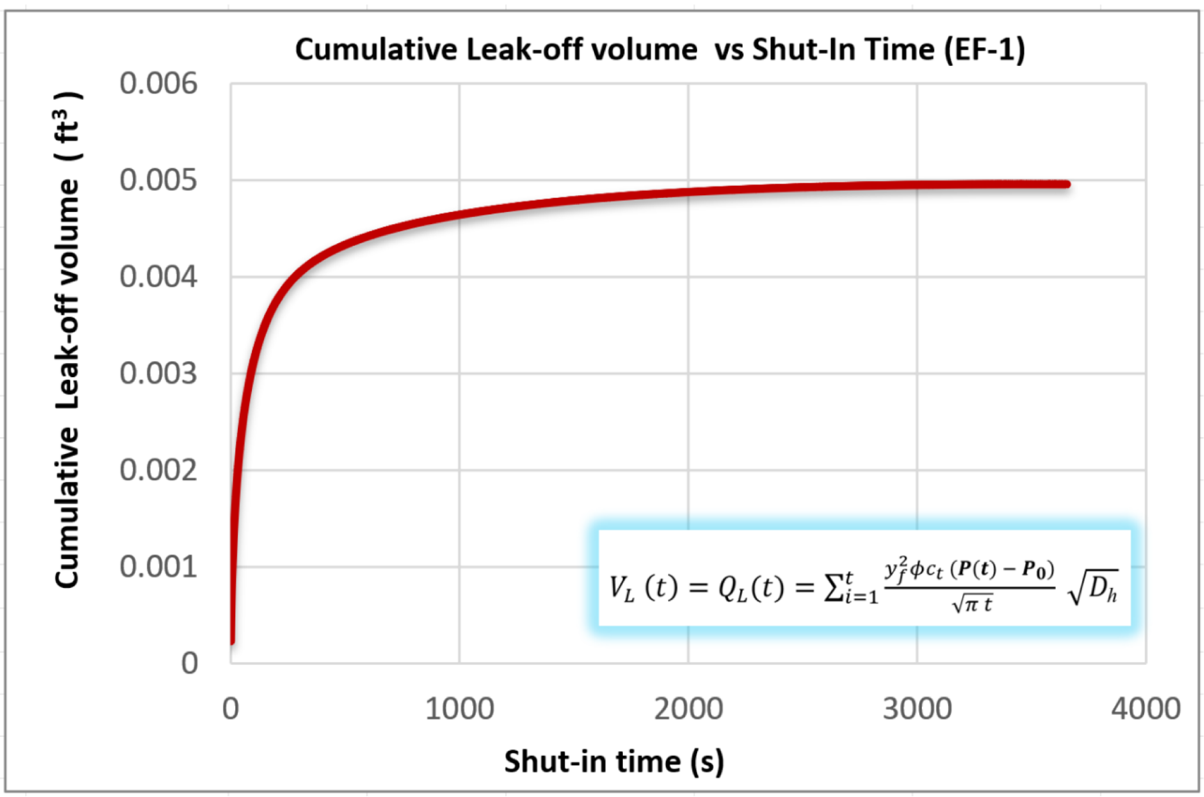


Figure 15—Cumulative leak-off volume obtained using the numerical summation approach Eq. (24)

Determining the pressure decline in the fracture

For the DFIT, we ultimately need have an analytical expression to match the observed pressure decline after shut-in Figure 8. The principle is given in Figure 13. The pressure in fracture at time of shut-in is P_f, and the analytical solution for the pressure-decline in the mini-fracture is:

$$P_{fall-off}(t) = P_f + \Delta P_{fall-off}(t) \tag{25}$$

Using fluid removal volume V_f due to leak-off we can compute the pressure loss due to fluid leak-off. First, we use the definition of the compressibility:

$$\frac{\Delta V_f(t)}{V_f} = -c_t \Delta P(t) \tag{26}$$

Solving for the $\Delta P(t)$ due to leak-off:

$$\Delta P_{fall-off}(t) = \Delta P(t) = -\frac{1}{C_t}\frac{\Delta V_f(t)}{V_f} = \frac{1}{C_t}\frac{V_L(t)}{V_f} \tag{27}$$

We know the value of V_f because it represents the total volume of fracturing fluid injected Table 3, and $V_L(t)$ was solved in Eq. (24).

One may now compute the pressure fall-off over time after inserting Eq. (27) into Eq. (25):

$$P_{fall-off}(t) = P_{f,intial} = -\frac{1}{C_t}\frac{\sum_{i=1}^{t}\frac{y_f^2 \phi c_t\left(\boldsymbol{P}(\boldsymbol{t}) - \boldsymbol{P_0}\right)}{\sqrt{\pi\, t}}\sqrt{D_h}}{V_f} \tag{28}$$

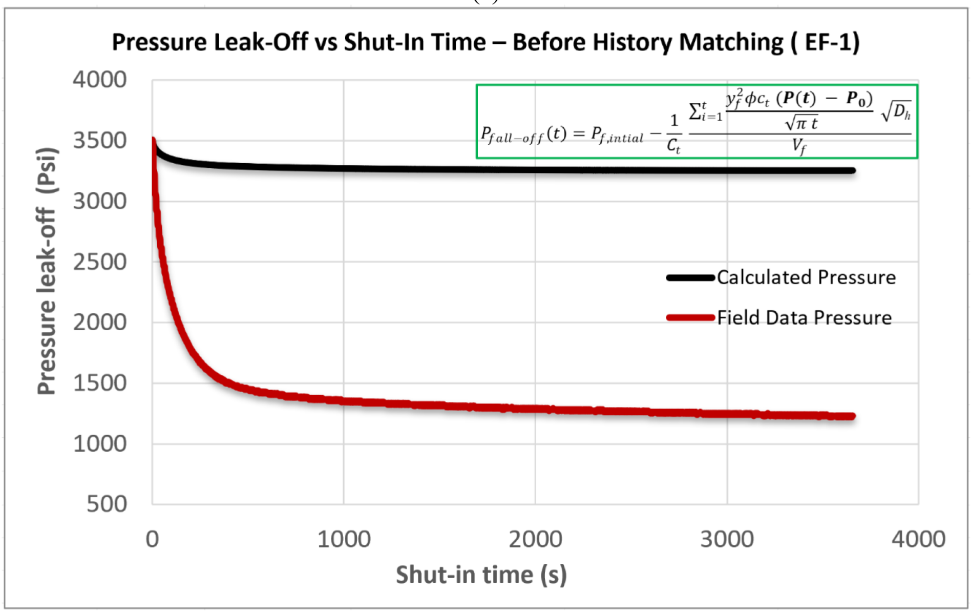
Table 8—Variable definitions for new fall-off pressure Eq. (28)

Symbol	Description	Source / Reference
$\boldsymbol{P_{f,intial}}$	The fracture propagation pressure at time of shut-in	Measured value **Table (7)**
$\boldsymbol{C_t}$	Total compressibility of the system	Weijermars (2022)
$\boldsymbol{y_f}$	Fracture half-length	Derived from modeling **Eq. (7)**
$\boldsymbol{\phi}$	Formation porosity	Weijermars (2022)
$\boldsymbol{\Delta P(t) = P\ (t) - P_0}$	Initial pressure difference	The principle in **Figure 13**
$\boldsymbol{D_h}$	Hydraulic diffusivity	Calculated from formation properties **Table (4)**
$\boldsymbol{t}$	Time after shut-in	Operational parameter
$\boldsymbol{V_f}$	injected fluid volume **Table 3**	Field data **Table (3)**
$\boldsymbol{V_L(t) = q_L(t)}$	Cumulative Leak-off volume over time	**Eq. (24)**

The solution for $V_L(t)$ to be used was given in Eq. (24) as the key Gaussian solution required for the DFIT history-matching. **Eq. (28)** therefore is a new equation for DFIT analysis, first derived in this study.

To illustrate the performance of the proposed model, Figure 16 compares the calculated pressure fall-off response with the actual field data for EF-1 well. **Subfigure (a)** presents the pressure decline before history matching, where the calculated response (black curve), obtained from Eq. (28) using the computed hydraulic diffusivity, shows a noticeable deviation from the field data (red curve). This mismatch indicates that the initial diffusivity assumption does not adequately capture the transient leak- off behavior observed in the reservoir. Subsequently, **Subfigure (b)** demonstrates the results after history matching, where the hydraulic diffusivity was adjusted through a least-squares history-matching procedure to minimize the difference between calculated and field pressures.

(a)



(b)

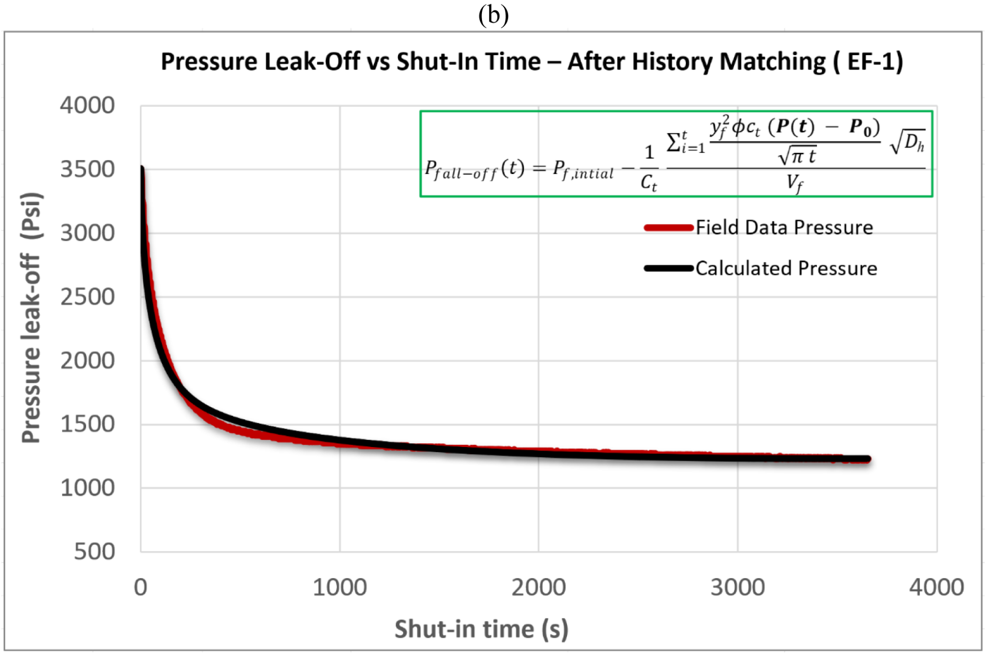


Figure 16—Pressure decline in the fracture after shut-in of pumps for EF-1: (a) before history matching using computed hydraulic diffusivity, and (b) after history matching using history-matched hydraulic diffusivity (Eq. 28).

In this case, the calculated response (black curve) aligns closely with the field data (red curve), confirming that the proposed Gaussian pressure transient method provides a reliable representation of pressure fall-off and leak-off behavior in unconventional reservoirs.

As part of the analysis, we also tested an analytical expression under the constant ΔP assumption. However, this approach did not provide a satisfactory match with the field data. The details of this case, including Eq. (28) and the corresponding plots, are documented in Appendix B.

History matching DFIT data for the seven wells

With V_f derived from Eq. (7), C_t from Weijermars (2022), and the cumulative leak-off volume $V_L(t)$ solved in Eq. (24), the pressure fall-off curves of Figure 8 were history-matched using Eq. (28) with a time-dependent ΔP to determine the hydraulic diffusivity. The least-squares fitting procedure was applied to minimize the error between calculated and measured fall-off responses.

The comparison between calculated and field data clearly shows the limitations of relying solely on the initially computed diffusivity and the improvement achieved after history matching. Figure 17 presents the leak-off profiles obtained using the computed hydraulic diffusivity values. In most cases, the calculated pressure decline (black curves) deviates from the field data (red curves), indicating that the initial diffusivity estimates do not fully capture the transient reservoir response. Figure 18 shows the corresponding results after history matching, where the hydraulic diffusivity was refined through a least- squares adjustment. In this case, the calculated curves align closely with the field data, confirming the robustness of the Gaussian pressure transient method in reproducing field behavior.

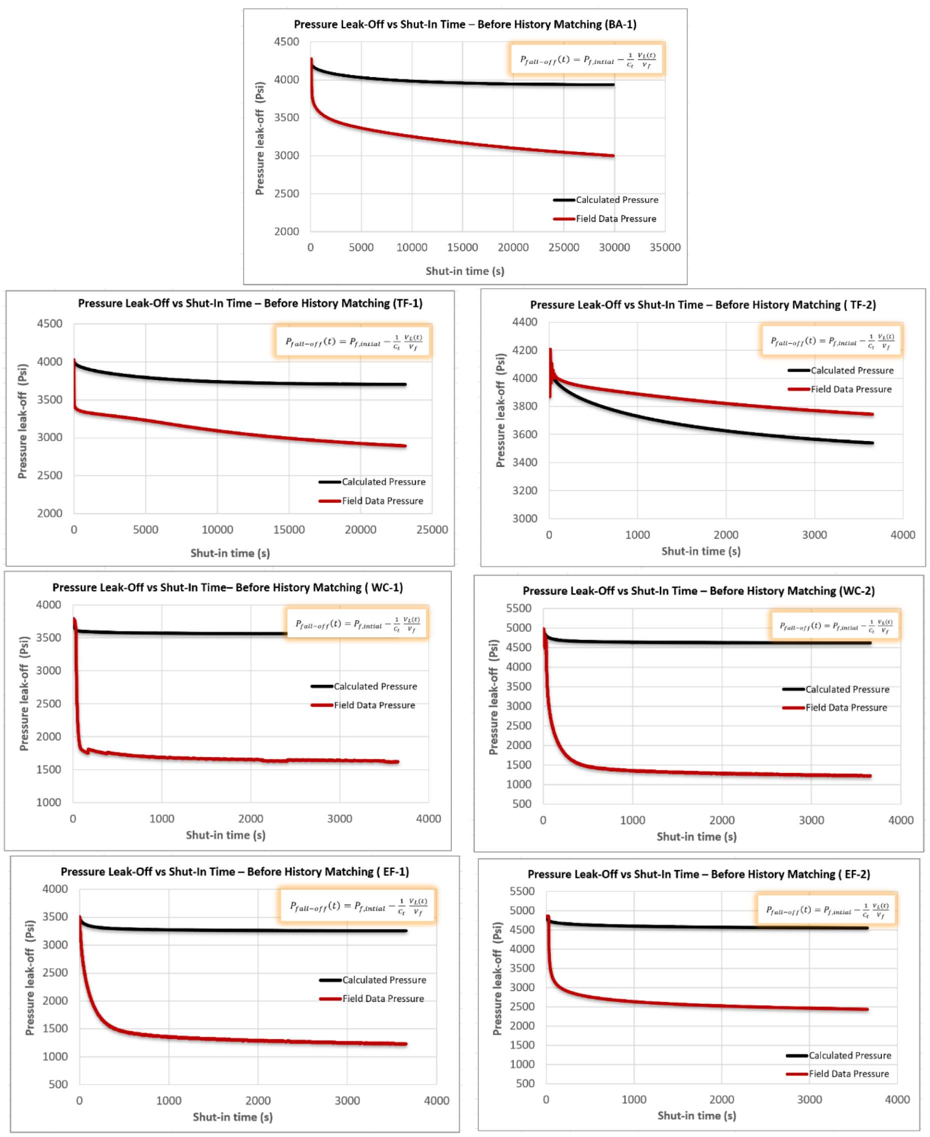


Figure 17—Leak-off profiles for each mini-frac test computed with hydraulic diffusivity estimation prior to the mini-frac test.

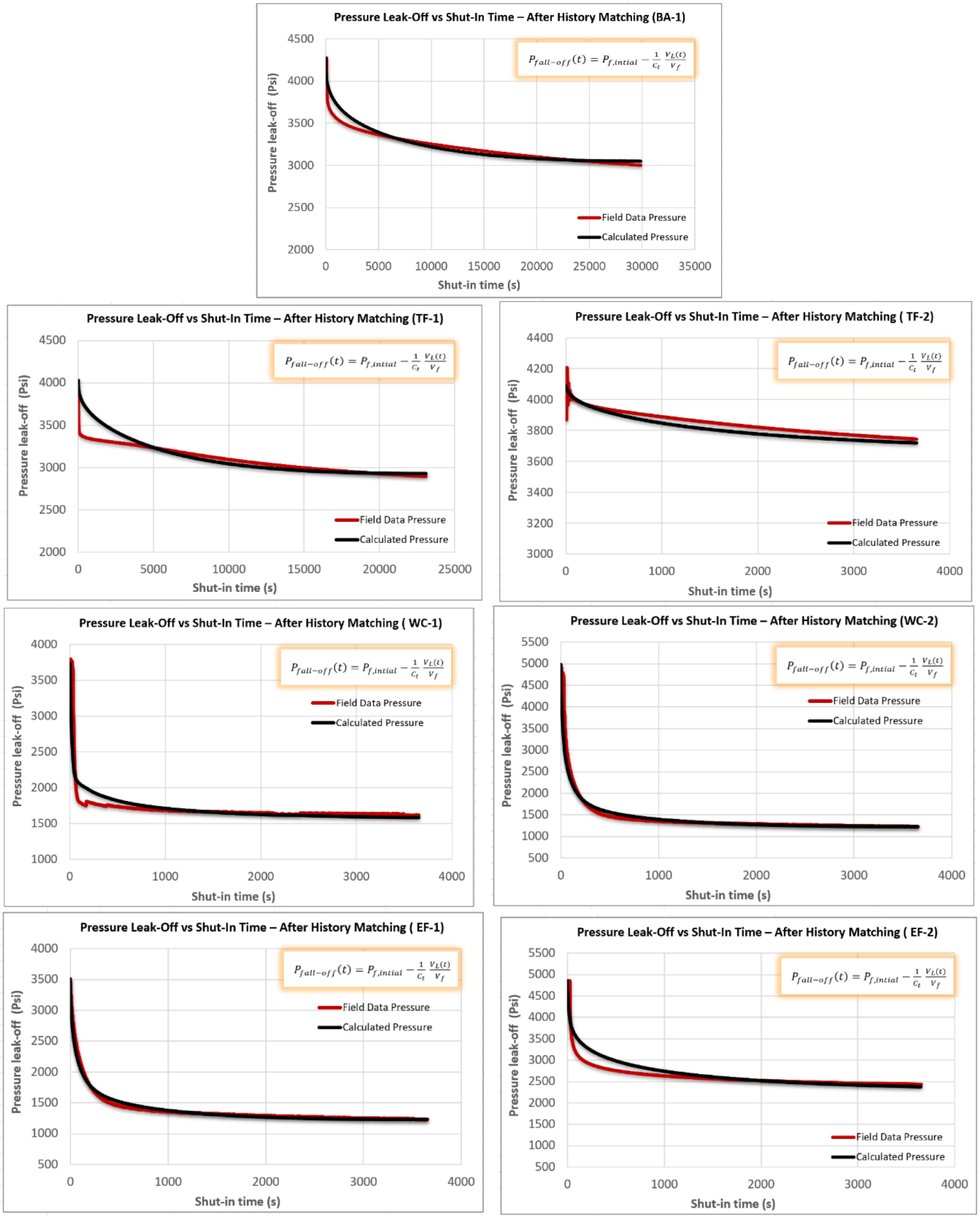


Figure 18—The leak-off profiles for each mini-frac test computed using history matched hydraulic diffusivity.

A quantitative summary of the results is provided in Table 9, which compares the computed and history-matched hydraulic diffusivity values for all seven wells across four shale plays. The table demonstrates that history matching consistently yields refined diffusivity estimates, reduces error in pressure decline modeling, and enhances the reliability of DFIT interpretation in unconventional reservoirs.

Table 9—Comparison of hydraulic diffusivity values

Shale Play	Well	Computed Hydraulic diffusivity D_h (ft²/s)	History Matched Hydraulic diffusivity D_h (ft²/s)
1	BA-1	0.0023	0.030
2	TF-1	0.0023	0.026
	TF-2	0.0114	0.005
3	WC-1	0.0011	0.178
	WC-2	0.0011	0.134
4	EF-1	0.0011	0.093
	EF-2	0.0011	0.079

Discussion

Challenges in conventional leak-off and permeability analysis

Applying the conventional DFIT analysis methods to unconventional reservoirs has revealed several challenges. Real DFIT data often show behaviors that violate the simplistic assumptions, leading to uncertainty or bias in interpreted parameters:

- ***Fracture Compliance and Gradual Closure:*** Traditional analysis assumes the fracture closes at a specific pressure and that fracture stiffness is constant until that point. In reality, fractures close progressively, the outer edges contact first, then closure proceeds inward as pressure declines. This means fracture compliance (the relationship of aperture with pressure) changes continuously during shut-in. If one assumes a sudden closure, the G-function or √t methods might pick a later point when the fracture is fully closed, underestimating the minimum stress. Studies have shown that ignoring varying compliance can cause underestimation of closure pressure by on the order of 100–200 psi. McClure et al. (2016) highlighted this by introducing a fracture-compliance method that picks an earlier "fracture contact" point (when walls first touch) on the G-function curve. The finding was that classical picks (tangent intersection) were often too low, due to continued fracture closure after the picked point. In short, non-ideal gradual closure behavior can lead to misidentification of $S_{h,min}$.
- ***Variable Leak-Off and Fluid-Loss Behavior:*** The leak-off rate during a DFIT can deviate from the ideal Carter model. Factors like pressure-dependent permeability (formation taking fluid faster as pressure is higher), the presence of natural fractures, or even changing fluid viscosity with pressure can all make leak-off non-linear and unsteady. Traditional G-function analysis assumes leak-off coefficient is constant; however, in shales the leak-off can slow down as the fracture pressure falls (or if filter cake builds up), violating the model. Barree et al. (2014) and others noted that the classical "normal leak-off" behavior is an oversimplification and can lead to significant errors in interpretation. For example, pressure-dependent leak-off (often called "height recession" or transverse storage effects) can cause the G-function derivative curve to misbehave (concave up/down shapes that don't match Nolte's type-curves). Furthermore, wellbore storage at early time

and fluid compressibility can mask true leak-off behavior initially, complicating the analysis of the first few minutes. All these make it challenging to apply one-size-fits-all analytical solutions for leak-off; each DFIT may require tailored interpretation beyond Nolte's standard plots (Gabry et al., 2024).

Typical DFIT test times in shale formations and the efficiency gain of GPT

Traditional well testing in ultra-tight shales is impractical because sustained flow without fractures cannot be achieved. DFIT has therefore become the diagnostic standard, but in practice DFIT tests in shale formations often require very long shut-in observation times to capture late-time pressure behavior. Our case studies confirm that test durations can extend from over 200 hours to nearly 500 hours (≈8–20 days) in the BA and TF wells, and still over 200 hours (≈8–10 days) in EF wells (Table 10). Such extended durations make DFIT operations costly and logistically challenging.

Table 10—Duration of DFIT tests for the seven study wells compared to GPT requirement

Well	DFIT Duration (Hours)	GPT Required Time (Hours)
BA-1	480	8.3
TF-1	214	6.4
TF-2	241	2.6
WC-1	1.6	1
WC-2	1.6	1
EF-1	200	1
EF-2	200	1.7

By contrast, the Gaussian Pressure Transient (GPT) workflow introduced in this study requires only a short portion of the pressure fall-off curve to extract all key reservoir parameters. As demonstrated in Figure 8, only a few hours of data are sufficient for GPT analysis, without compromising accuracy. This represents a major gain in operational efficiency: DFIT operations that historically consumed weeks of observation time can now be completed in hours, drastically reducing both cost and operational downtime.

Conclusions

Advantages of the proposed GPT method

- **Captures gradual pressure decline accurately**
  Traditional models assume sudden fracture closure. GPT models the pressure as a smooth diffusion process; better matching real data where closure is gradual and non-linear.
- **No need for subjective "closure pressure"**
  Unlike G-function or √t methods, GPT doesn't require you to guess the fracture closure point. The model continuously fits the pressure fall-off, avoiding uncertainty and human bias.
- **Avoids rigid leak-off assumptions (e.g., carter model)**
  GPT doesn't assume a √t leak-off profile. Instead, it uses physics-based pressure gradients to derive leak-off dynamically from the actual pressure profile.

- **Provides direct estimates of fracture geometry**
  By integrating Sneddon's fracture theory and volume balance, the GPT approach computes half-length and aperture without separate fracture modeling tools.
- **Enables fast, spreadsheet-based field analysis**
  No complex simulation software is needed. GPT solutions are algebraic and can be rapidly applied in operational settings; ideal for time-sensitive field diagnostics.
- **More consistent parameter estimation across wells**
  GPT-derived parameters (like diffusivity and permeability) are grounded in a consistent framework, making cross-well comparison more reliable than classical methods.

Summary of key achievements and results

- Developed and applied a Gaussian-based analytical method for DFIT interpretation.
- Analyzed 7 wells from 4 major shale formations (Bakken, Three Forks, Wolfcamp, Eagle Ford).
- Introduced a new equation for pressure fall-off using leak-off volume.
- Demonstrated the limitations of constant ΔP models.
- time-dependent ΔP models matched field data more accurately.
- Accurately predicted leak-off rate, cumulative volume, and pressure fall-off.
- Final pressure curves matched real well data across all scenarios.
- Hydraulic diffusivity was refined for each well, improving reservoir understanding.

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Acronyms

DFIT	Diagnostic Fracture Injection Test
GPT	Gaussian Pressure Transient
ISIP	Instantaneous Shut-In Pressure
SITP	Shut-In Tubing Pressure
FPP	Fracture Propagation Pressure
ACA	After-Closure Analysis
BCA	Before-Closure Analysis
FBA	Flowback Analysis
RU	Rig Up
RD	Rig Down
RHOS	Rods Hot Oil systems
BBL	Barrels
BPM	Barrels Per Minute
WC	Wolfcamp (well designation)
EF	Eagle Ford (well designation)
TF	Three Forks (well designation)
BA	Bakken (well designation)

Nomenclature

P_f	Fracture pressure at shut-in
P_0	Initial reservoir pressure
ΔP *or* $\Delta P(t)$	Pressure difference (constant or time-dependent)
y_f	Fracture half-length
w_f	Fracture width (aperture)
h_f	Fracture height
α_f	Fracture half-width
V_f	Total fracture volume
$V_L(t) = Q(t)$	Cumulative leak-off volume
$q(t)$	leak-off rate
$P_{fall\text{-}off}(t)$	Fracture pressure decline during shut-in
$\nabla P(t)$	Pressure gradient on fracture surface
μ	Fluid viscosity
ϕ	Porosity
c_r	Rock compressibility
c_f	Fluid compressibility
c_t	Total system compressibility
k	Permeability
D_h	Hydraulic diffusivity
E	Young's modulus
v	Poisson's ratio
A	Fracture surface area
x	Distance from fracture face
erf ()	Error function

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Appendix A

Workflow steps for this study

The principal workflow steps developed during this study are briefly outline below.

Data collection

- **Source DFIT data**: Gather pressure fall-off data from 7 wells across four shale formations (Bakken, Three Forks, Wolfcamp, Eagle Ford).
- **Input parameters:**
  - Surface-measured fracture pressure ($P_{F@Surface}$).
  - Hydrostatic pressure ($P_{hydrostatic} = 0.052 \times \rho \times h$).
  - Minimum in-situ stress ($\sigma 3$) from breakout tests.
  - Injected fluid volume (V_f) from pump logs.
  - Elastic rock properties (E, v) from sonic logs and core tests.
  - Reservoir parameters (ϕ, μ, c_t) from literature (Weijermars, 2022).

Data preprocessing

- **Calculate bottom-hole pressure:**

$$P_{F@downhole} = P_{F@Surface} + P_{hydrostatic} \quad \text{(A.1)}$$

- **Compute Sneddon pressure (P_S):**

$$P_S = P_{F@downhole} - \sigma 3 \quad \text{(A.2)}$$

- **Convert units**: Ensure consistency (e.g., V_f from bbl to ft^3).

Fracture growth modeling

- **Sneddon's solution:** Calculate fracture half-length (y_f) using:

$$y_f = \left(\frac{4E}{\pi(1 - v^2)} \frac{V_f}{P_S} \right)^{1/3} \quad \text{(A.3)}$$

- **Assumptions:**
  - Fracture height (h_f) grows at half the rate of length ($\frac{y_f}{h_f} = 4$).
  - Elliptical fracture geometry (plane strain).

Pressure transient analysis

- **Gaussian pressure model:**

$$P_{(x,t)} = P_0 + \Delta P_{(x,t)} = P_0 + \Delta P_0 \left(1 - erf\left(\frac{x}{2\sqrt{D_h t}} \right) \right) \quad \text{(A.4)}$$

- **Hydraulic diffusivity (D_h):**

$$D_h = \frac{k}{\phi \mu c_t} \tag{A.5}$$

Leak-off quantification

- **Leak-off rate $q_L(t)$:**

$$q_L(t) = -\frac{y_f^2 \phi c_t \Delta P}{\sqrt{\pi\, t}} \sqrt{D_h} \tag{A.6}$$

- **Cumulative leak-off $V_L(t)$:**

$$V_L(t) = Q_L(t) = -\frac{y_f^2 \phi c_t \left(P(t) - P_0\right)}{\sqrt{\pi\, t}} \sqrt{D_h} \tag{A.7}$$

- **Pressure decline in the fracture $P_{\text{fall-off}}(t)$**

$$P_{fall-off}(t) P_{f,intial} - \frac{1}{C_t} \frac{\sum_{i=1}^{t} q_{L,i}}{V_f} \tag{A.8}$$

$$P_{fall-off}(t) P_{f,intial} - \frac{1}{C_t} \frac{\left(\frac{y_f^2 \phi c_t \left(P(t) - P_0\right)}{\sqrt{\pi\, t}} \sqrt{D_h} \right)}{V_f} \tag{A.9}$$

History matching and validation

- **Validation**: Match theoretical curves with field data Figure 8.
- **Compare D_h** (Table 9).
- **Output**: Adjusted D_h.

Conclusions

- **Key results:**
  - Fracture dimensions (y_f) for each well.
  - Leak-off rates and pressure decline profiles.
  - Hydraulic diffusivity values for each shale formations.
- **Advancements:**
  - Gaussian model better captures low-permeability shale transients, and captures complex pressure transients.

Appendix B

Analytical solution attempt for cumulative leak-off and pressure decline

Determining the cumulative leak-off volume for a fracture

Before the numerical summation approach presented in the main text, an analytical solution for the cumulative leak-off volume was also tested.

Starting from integrates the leak-off rate (Eq. 22), the cumulative volume was defined as:

$$V_L(t) = Q_L(t) = \int_{t=0}^{n} q_L(t).dt \tag{B.1}$$

This leads to the closed form analytical expression:

$$V_L(t) = Q_L(t) = -\frac{2y_f^2 \phi c_t \, \Delta P_0}{\sqrt{\pi}} \sqrt{D_h t} \tag{B.2}$$

When $\Delta P(t) = P(t) - P_0$ was substituted into Eq. D1,

$$V_L(t) = Q_L(t) = -\frac{2y_f^2 \phi c_t \left(P_{(t)} - P_0\right)}{\sqrt{\pi}} \sqrt{D_h t} \tag{B.3}$$

The results became non-physical, with cumulative volume trends showing unrealistic behavior (Figure B.1).

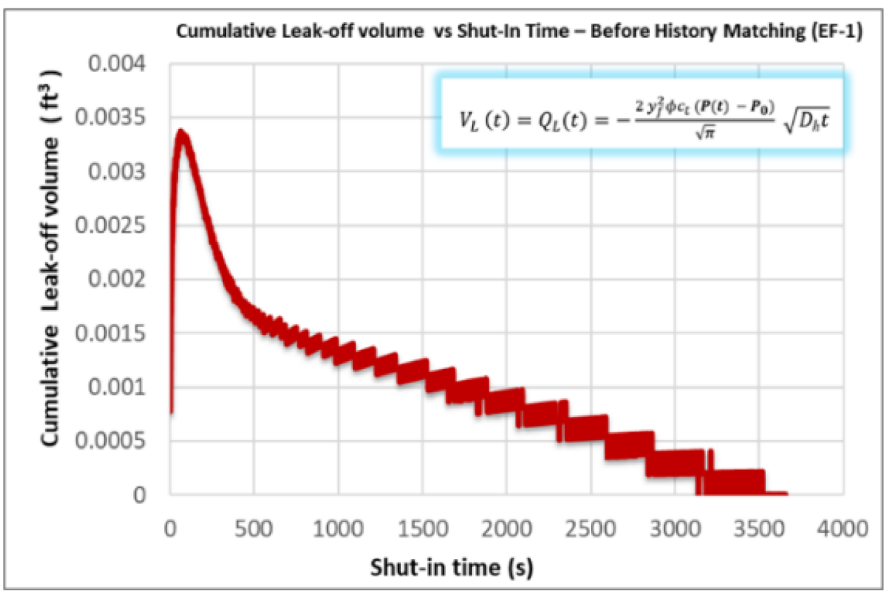


Figure B.1—Cumulative leak-off volume with Δ*P(t)* (analytical attempt, failed)

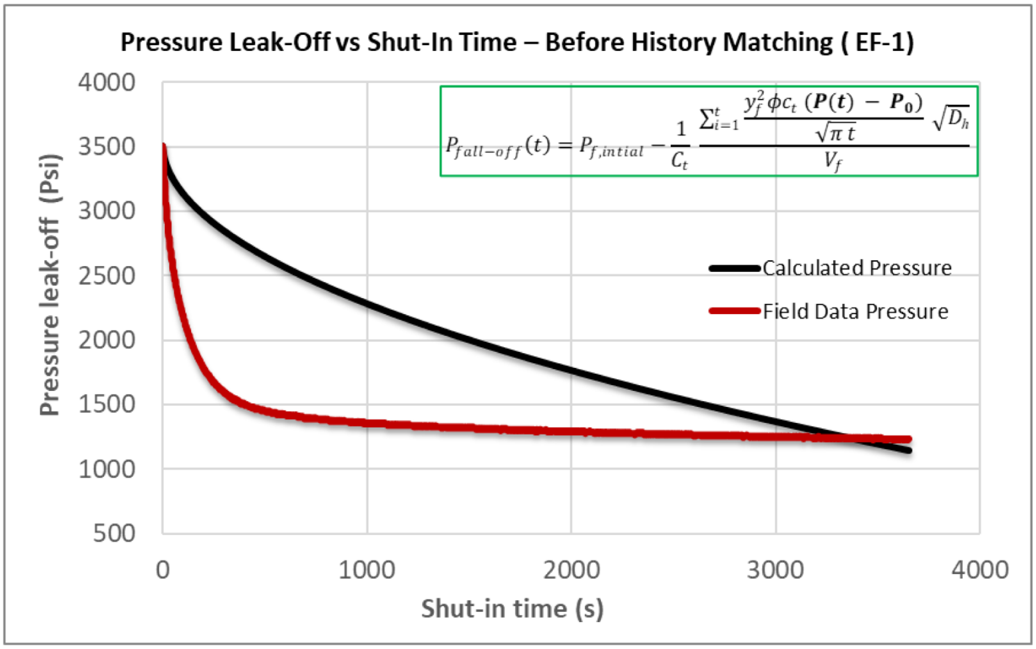
Determining the pressure decline in the fracture

The analytical solution for $V_L(t)$ to be used was given in Eq. (B.3), as the key Gaussian solution required for the DFIT history-matching.

$$P_{fall-off}(t) = P_{f,intial} - \frac{1}{C_t} \frac{\left(\dfrac{2y_f^2 \phi c_t \left(P_f - P_0\right)}{\sqrt{\pi}} \sqrt{D_h t}\right)}{V_f} \tag{B.4}$$

Although Eq. (B.4) offers a new analytical expression for pressure decline during early-time leak-off under constant fracture pressure assumptions, the model did not yield a satisfactory match with the actual field data, as shown in Figure B.2. Specifically, the calculated pressure decline curve deviates noticeably from the observed field pressure, indicating that the model does not fully capture the reservoir's transient response under constant fracture pressure case.

(a)



(b)

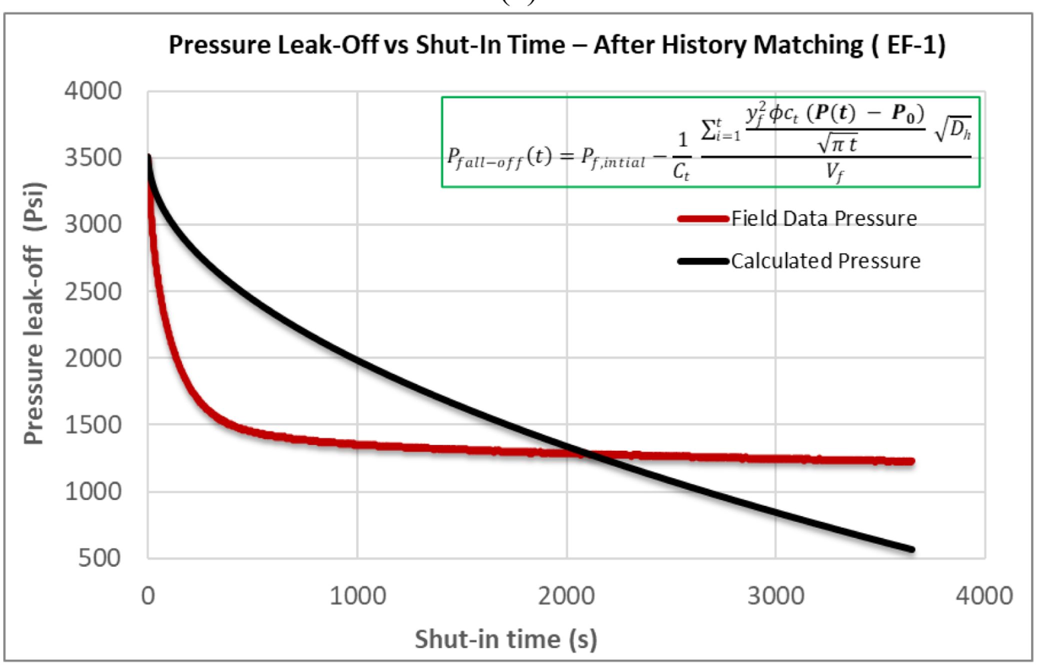


Figure B.2—Pressure decline in the fracture after shut-in of pumps under constant Δ*P* assumption. (a) before history matching using computed hydraulic diffusivity, and (b) after history matching using history-matched hydraulic diffusivity (Eq. B.4).

Fracture pressure declines continuously after shut-in, making ΔP time-dependent. The analytical model fails to accommodate this dynamic behavior, particularly in ultra-low-permeability formations where pressure propagation and leak-off are highly sensitive to diffusivity and compressibility variations over time. This analytical limitation will be revisited in future work, where we aim to further develop or adjust the Gaussian solution Eq. (B.4) to improve its applicability under non-ideal and dynamic field conditions. To address this limitation, a numerical summation method Eq. (28) was introduced as a more flexible alternative. This approach accounts for the time-dependent nature of ΔP and offers improved accuracy in cumulative leak-off and pressure decline predictions.